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(54) **AUTONOMOUS TRACTOR USING
COUNTER FLOW-DRIVEN PROPULSION**

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See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

5,184,676 A 2/1993 Graham et al.
5,392,715 A 2/1995 Pelrine
(Continued)

FOREIGN PATENT DOCUMENTS

WO 9802634 1/1988
WO 2016076875 5/2016

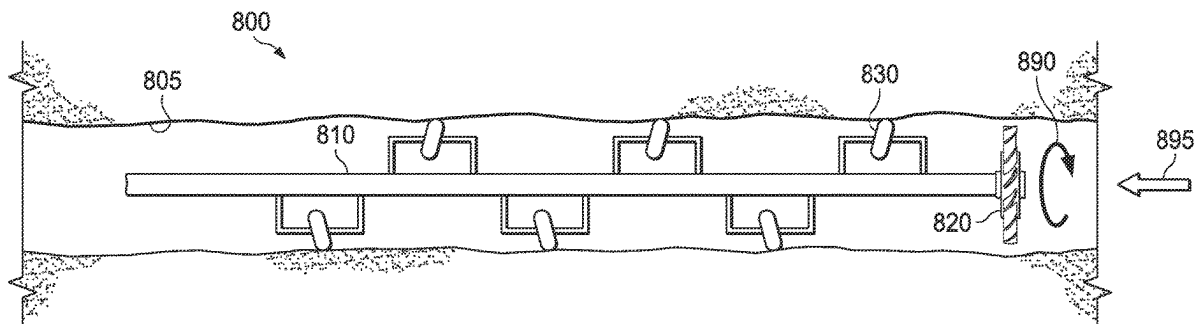
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(57) **ABSTRACT**

Provided is a wellbore tractor and method for operating a
well system. The wellbore tractor, in one aspect, includes a
base member, and one or more turbines fixed to the base
member for rotating the base member in a first rotational
direction based upon a first direction of fluid flow. The
wellbore tractor, according to this aspect, further includes
one or more wellbore engaging devices radially extending
from the base member, the one or more wellbore engaging
devices contactable with a surface of a wellbore for displac-
ing the base member and one or more turbines axially
downhole as the one or more turbines rotate in the first
rotational direction.

17 Claims, 12 Drawing Sheets



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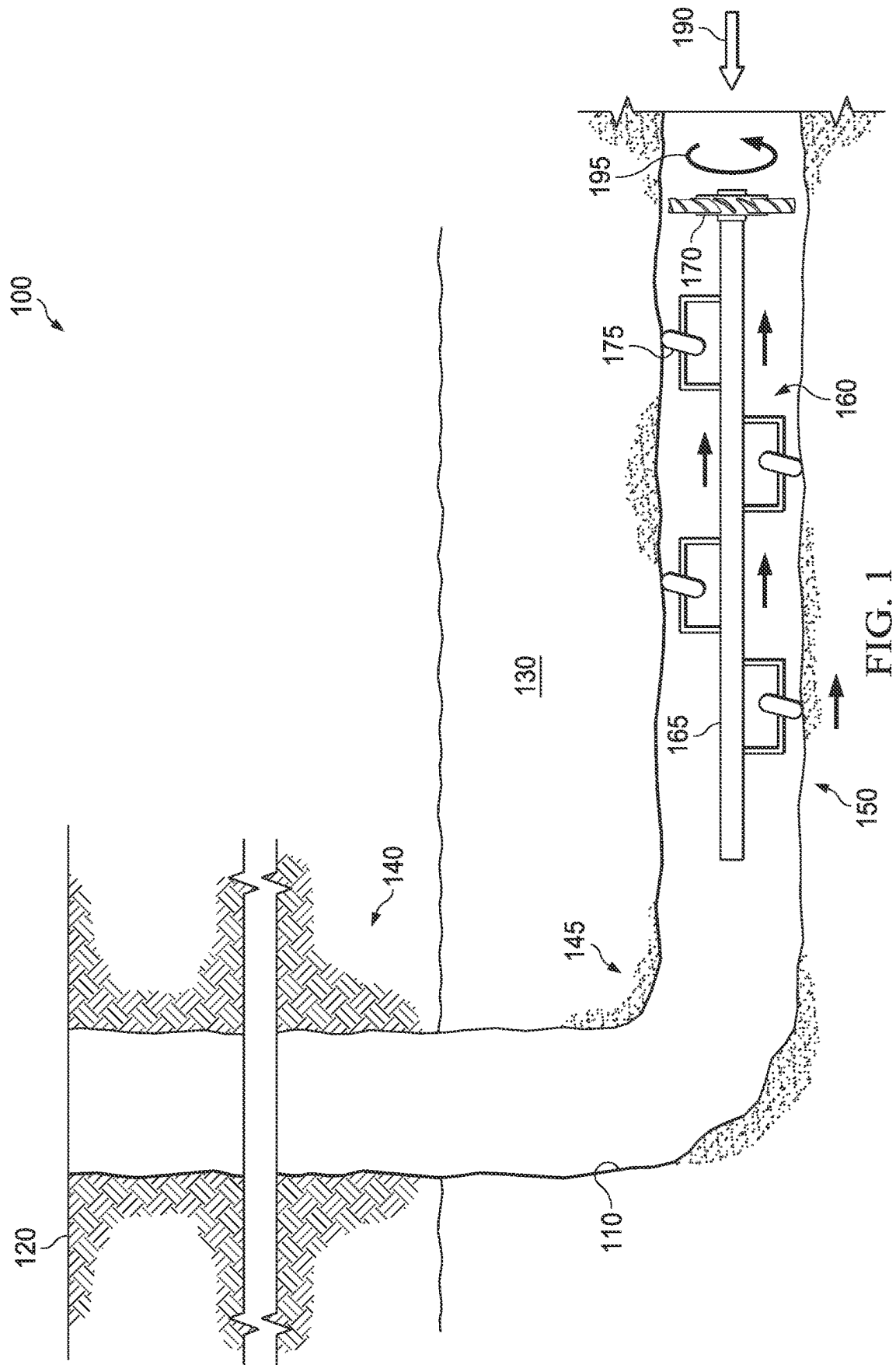
E21B 47/07 (2012.01)

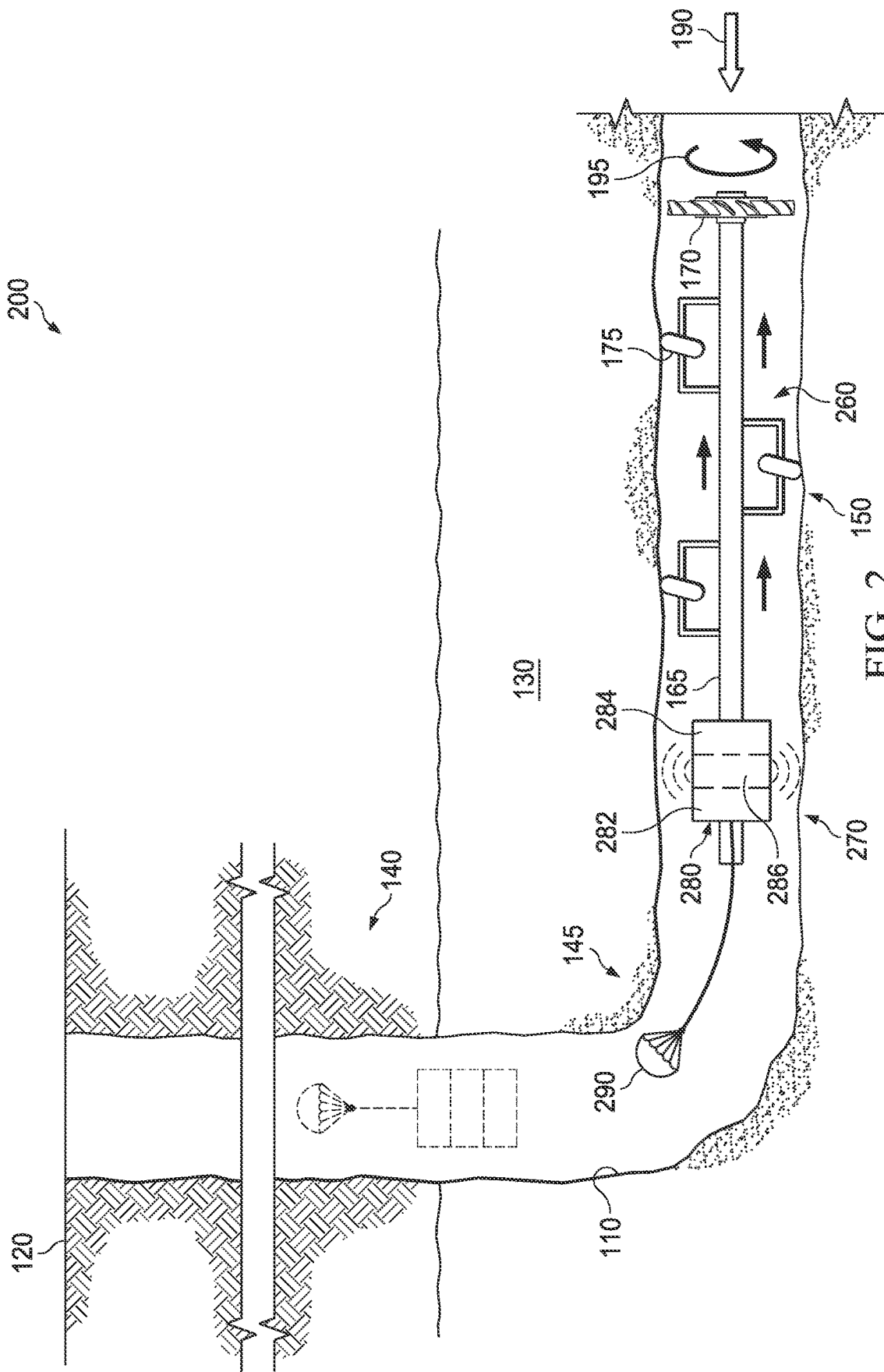
(56) **References Cited**

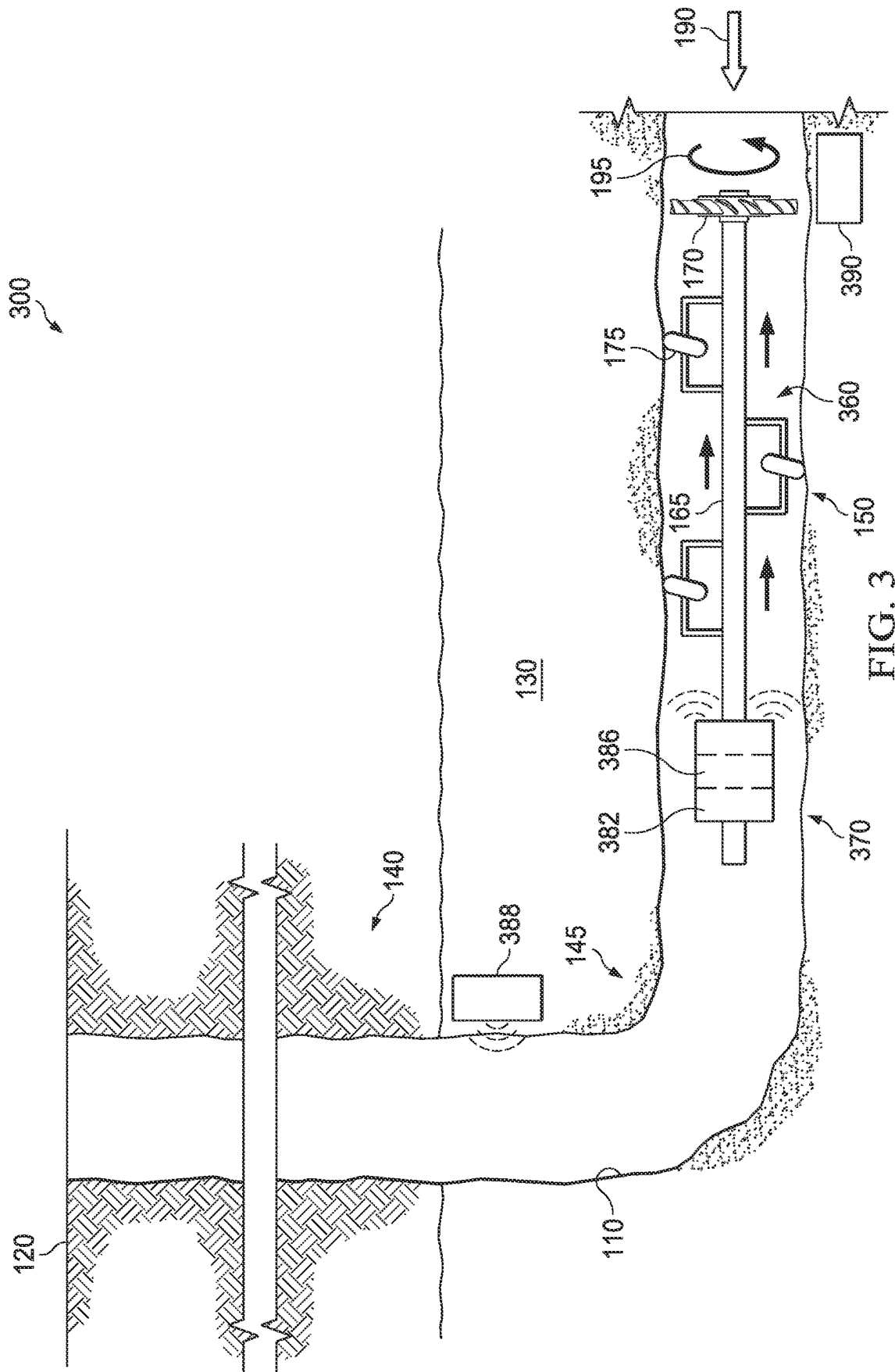
U.S. PATENT DOCUMENTS

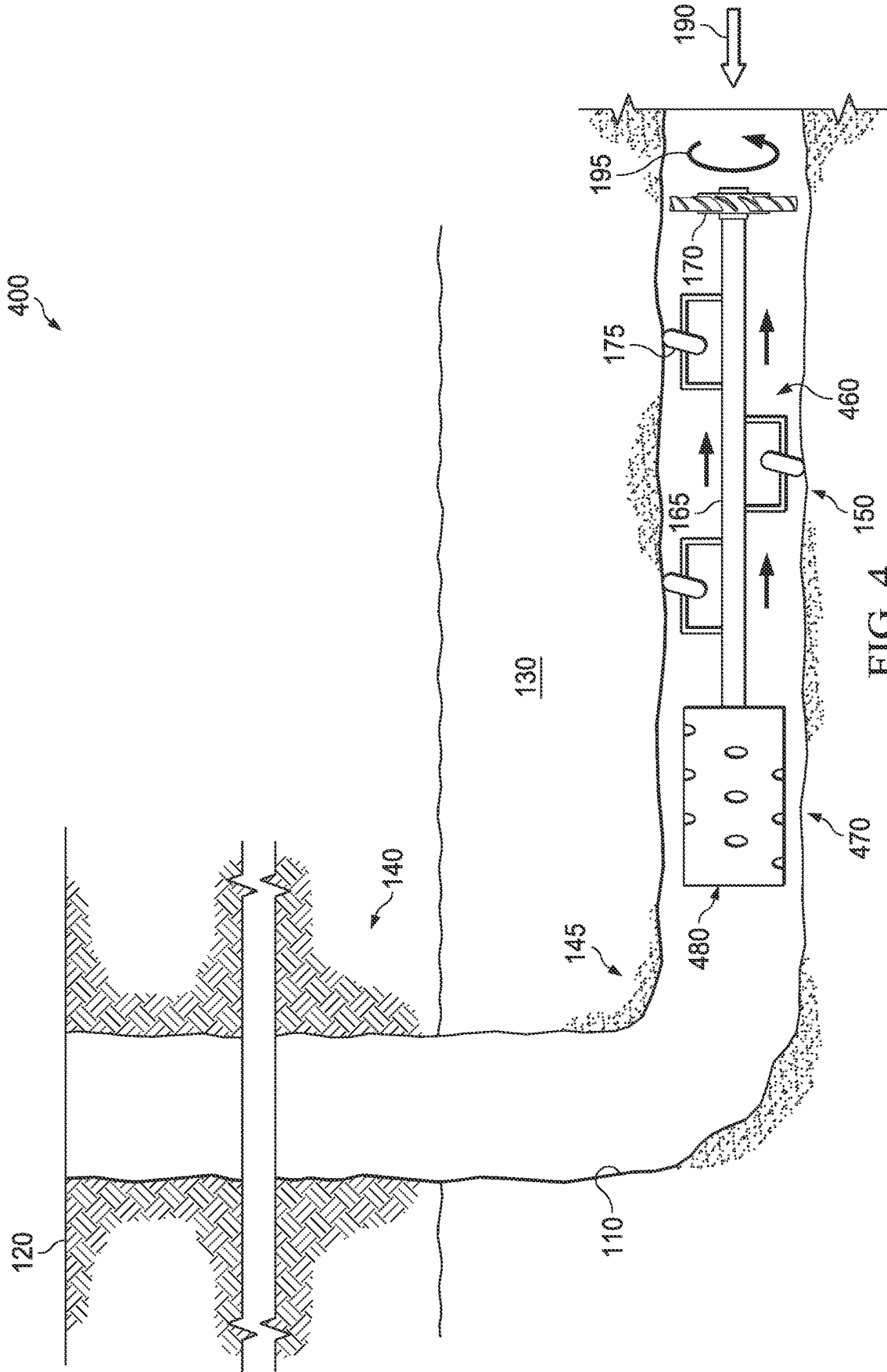
6,378,627	B1	4/2002	Tubel et al.	
6,722,442	B2 *	4/2004	Simpson	F16L 55/26 166/173
6,843,317	B2	1/2005	Mackenzie	
2004/0055746	A1	3/2004	Ross et al.	
2005/0016302	A1 *	1/2005	Simpson	E21B 47/00 73/865.8
2005/0241824	A1 *	11/2005	Burris, II	E21B 47/04 166/134
2005/0288819	A1	12/2005	de Guzman	
2008/0053663	A1	3/2008	Moore	
2011/0035152	A1	2/2011	Durocher et al.	
2013/0240197	A1	9/2013	Hallundbaek et al.	
2015/0252677	A1 *	9/2015	Scholz	E21B 41/0085 29/889.71
2016/0251941	A1	9/2016	Murphree et al.	

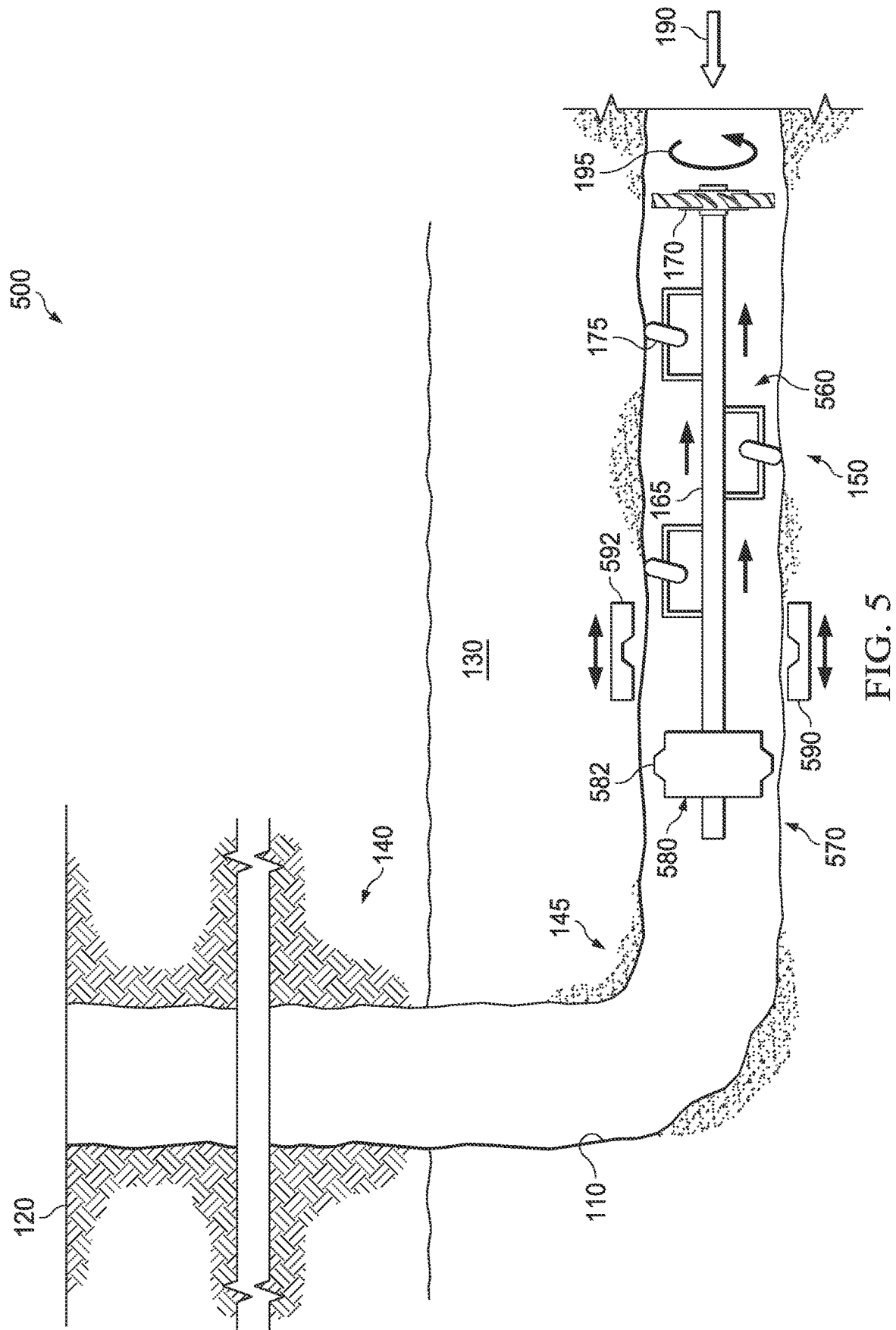
* cited by examiner

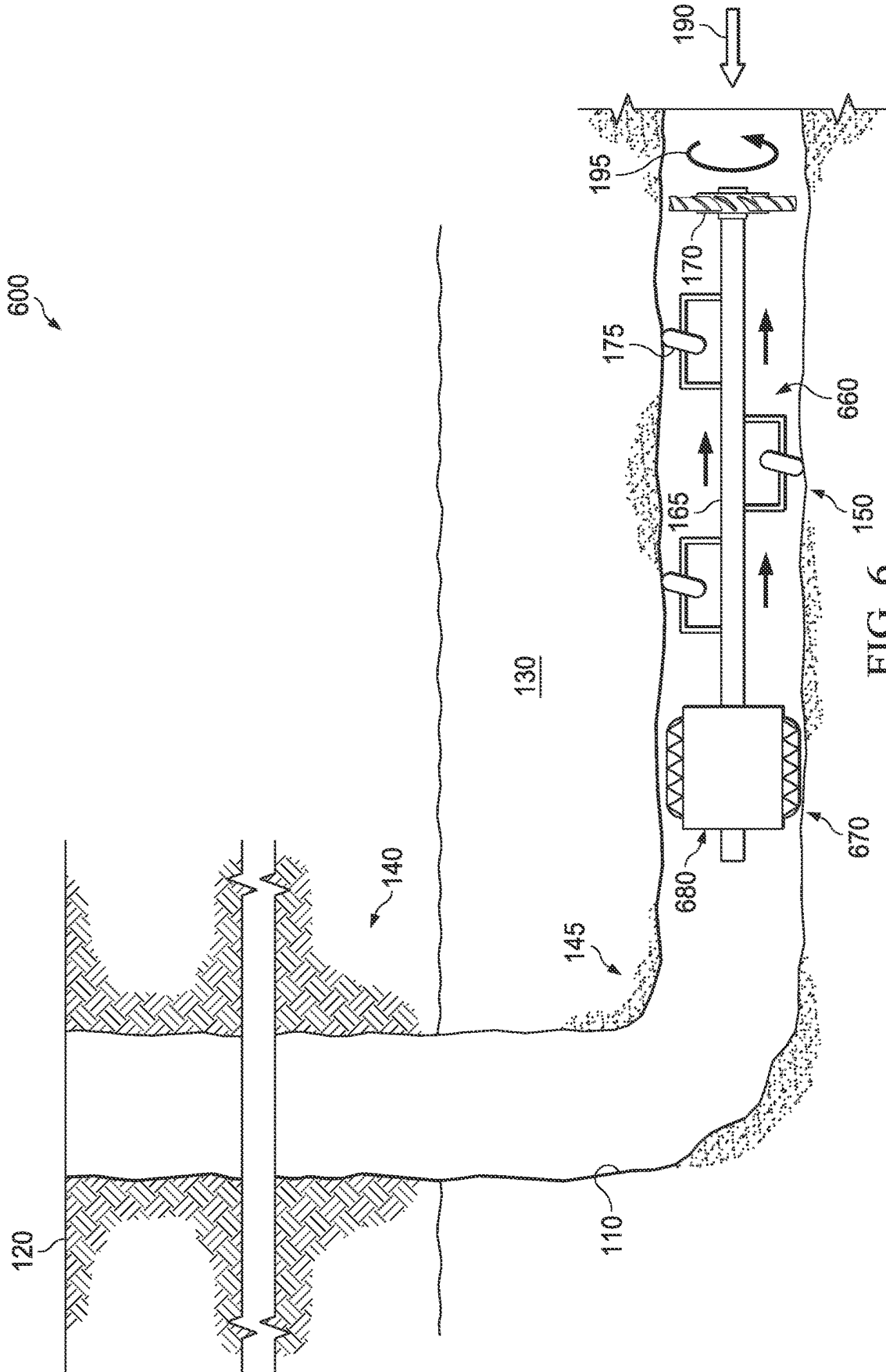












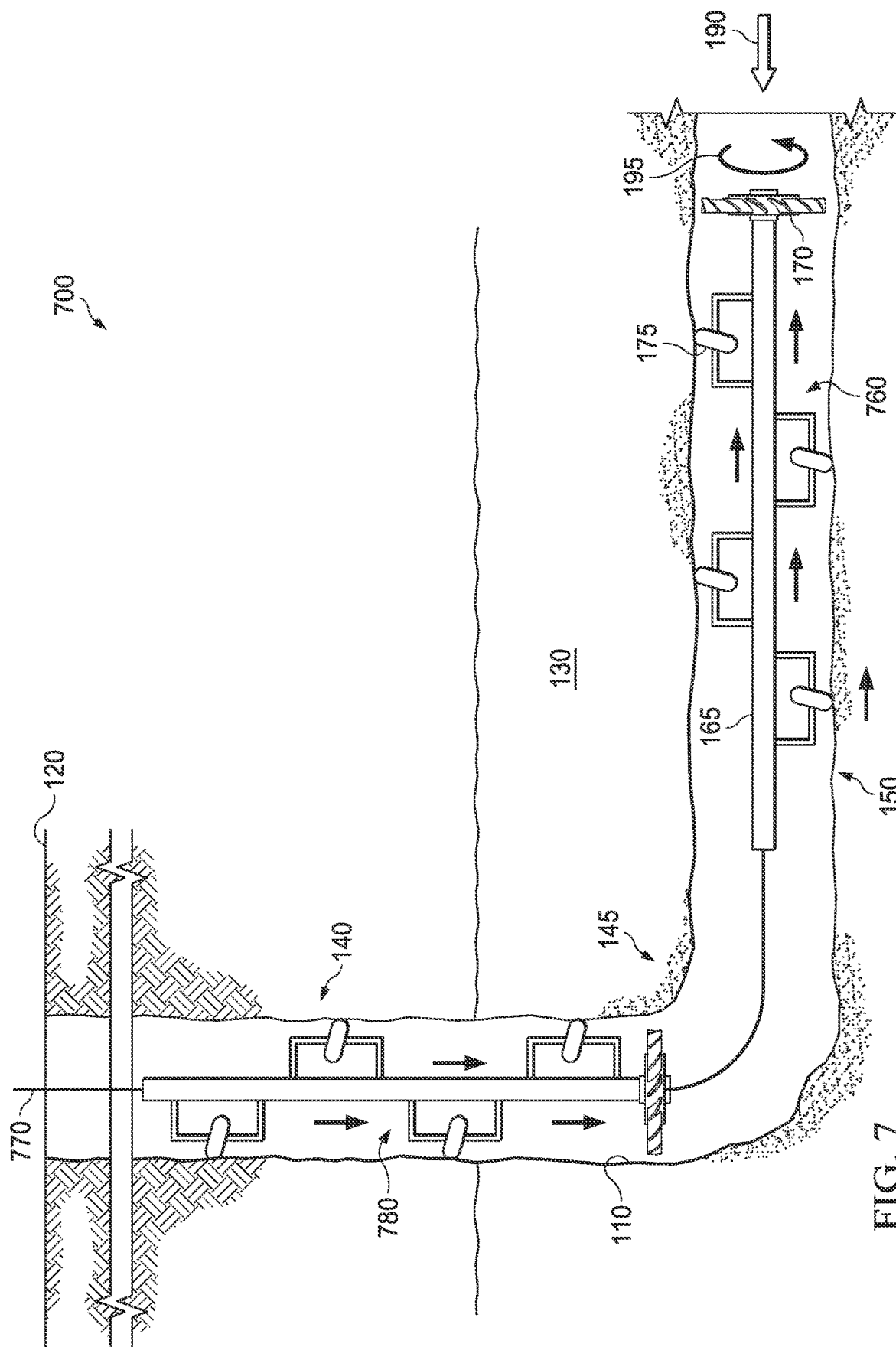
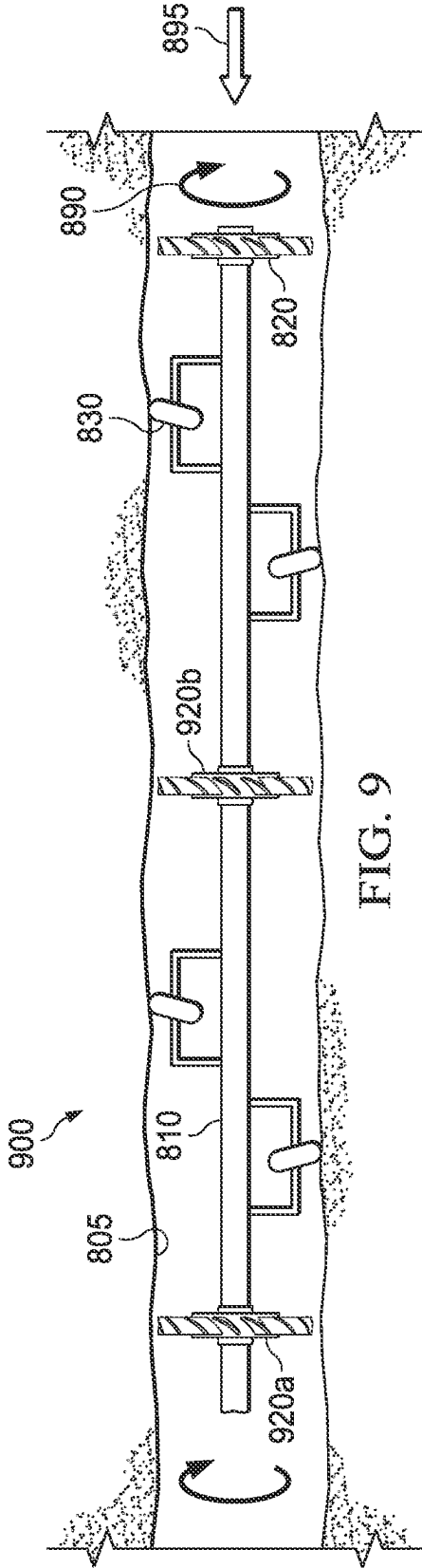
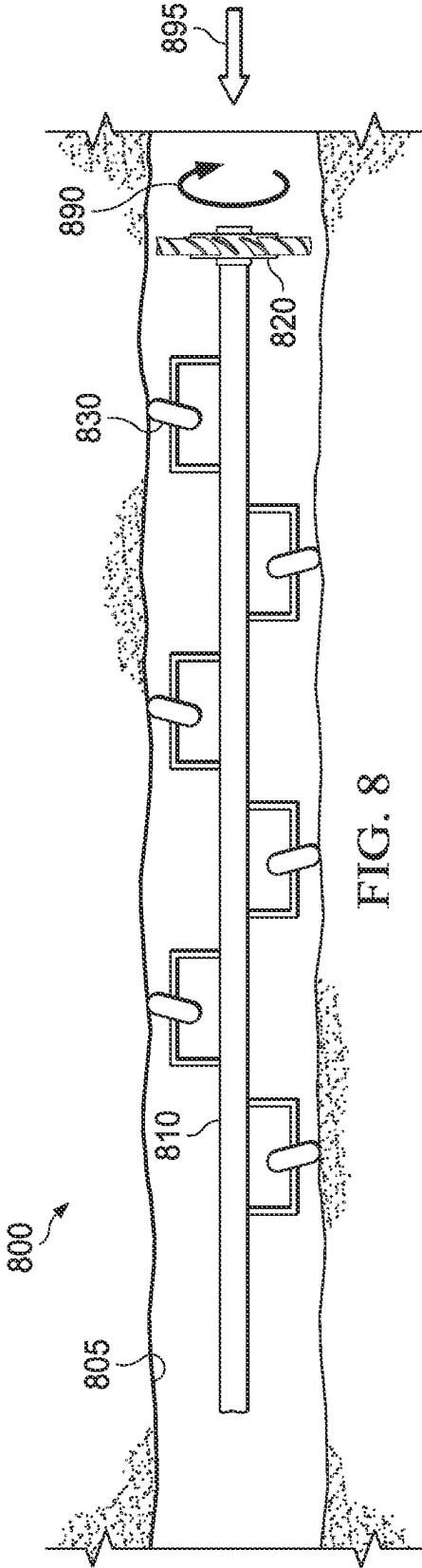
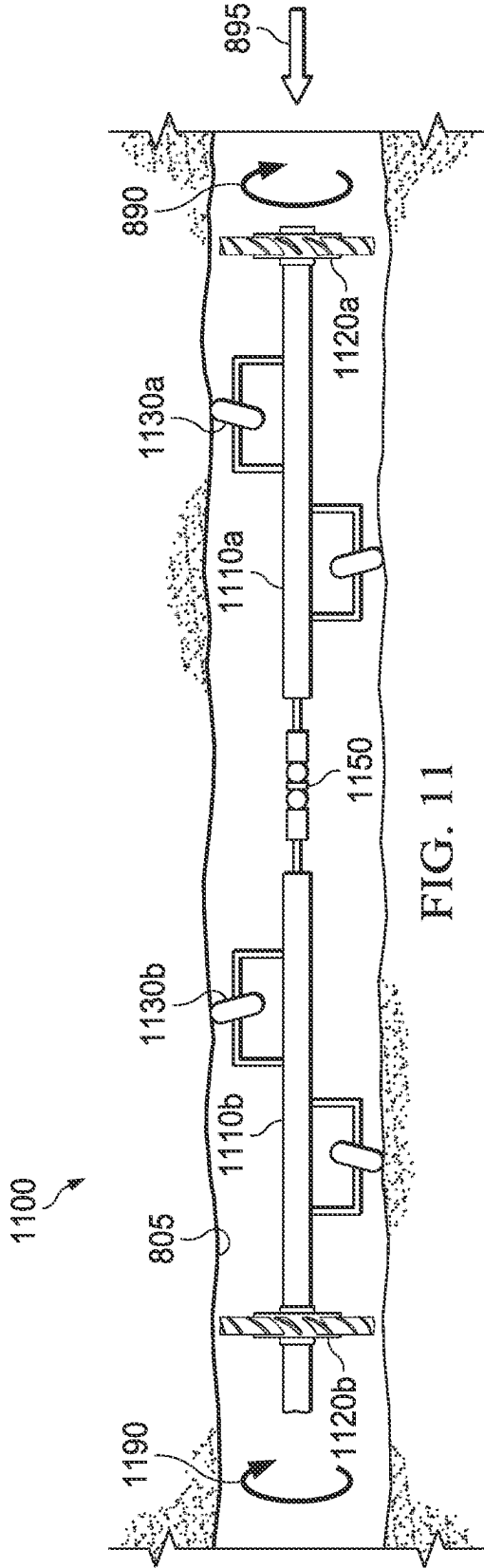
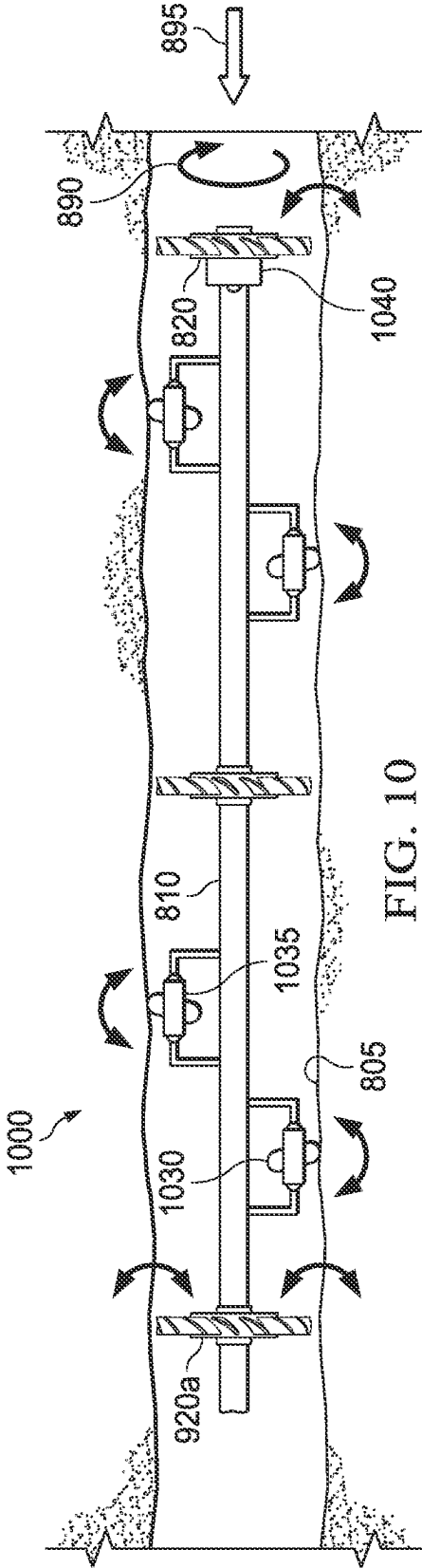
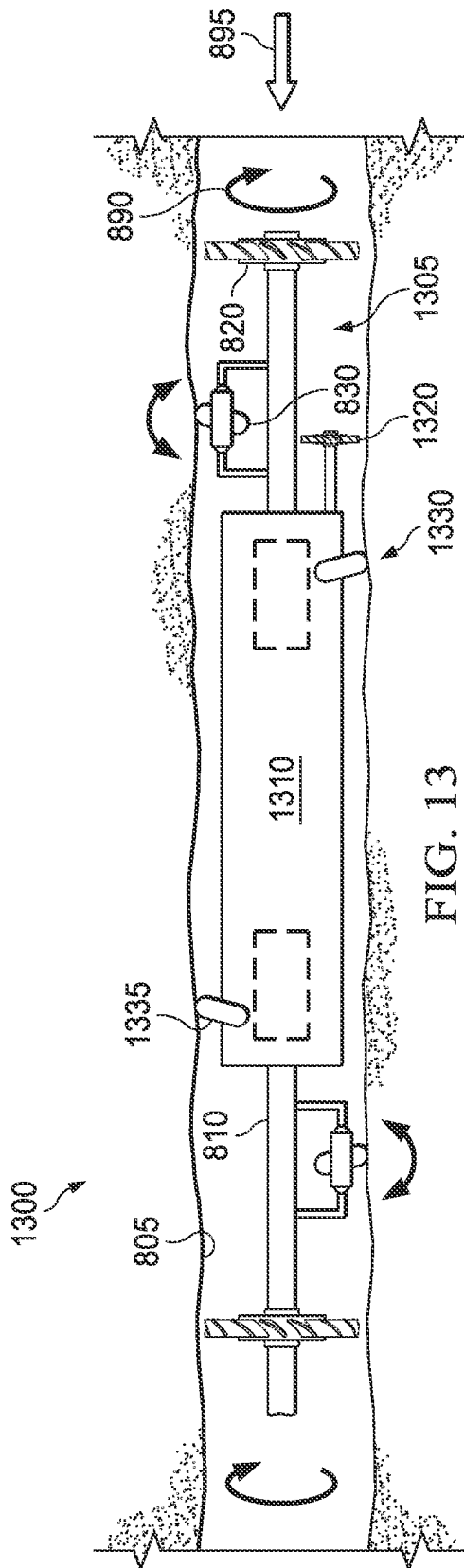
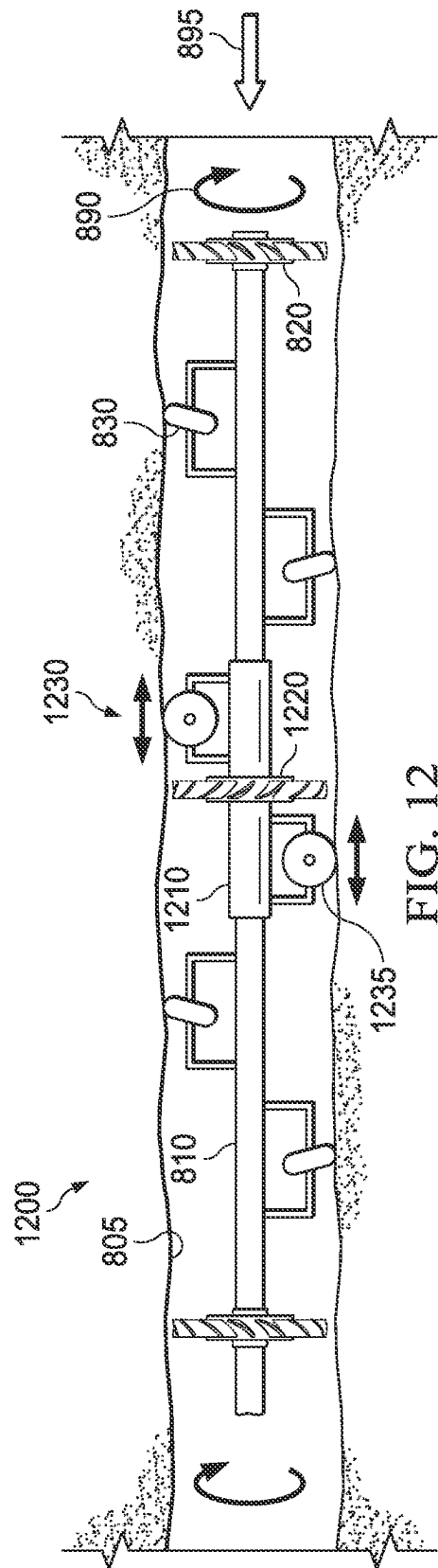
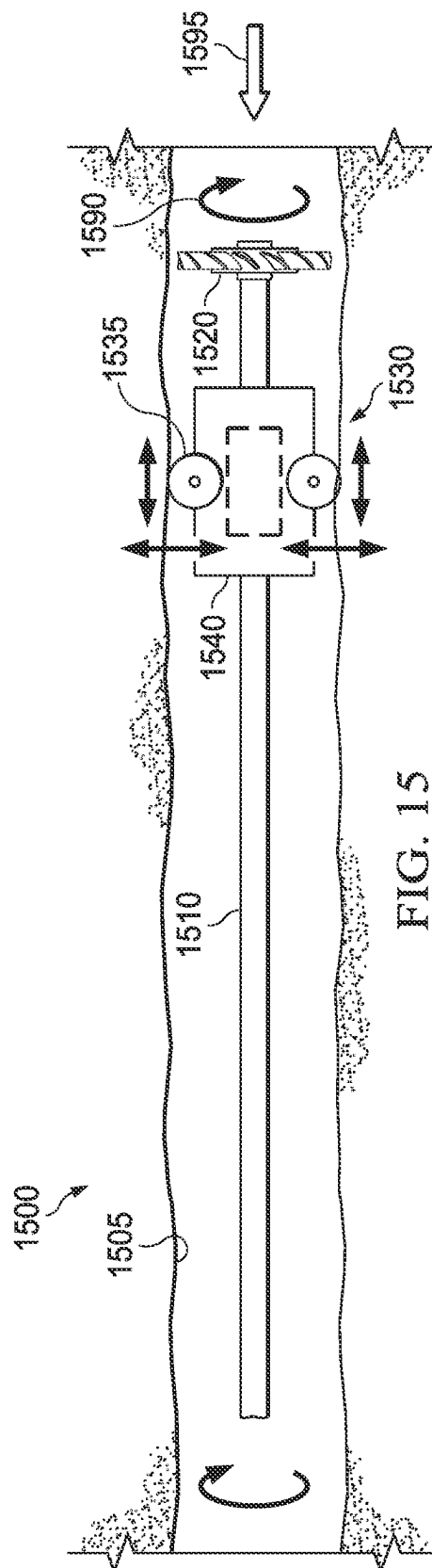
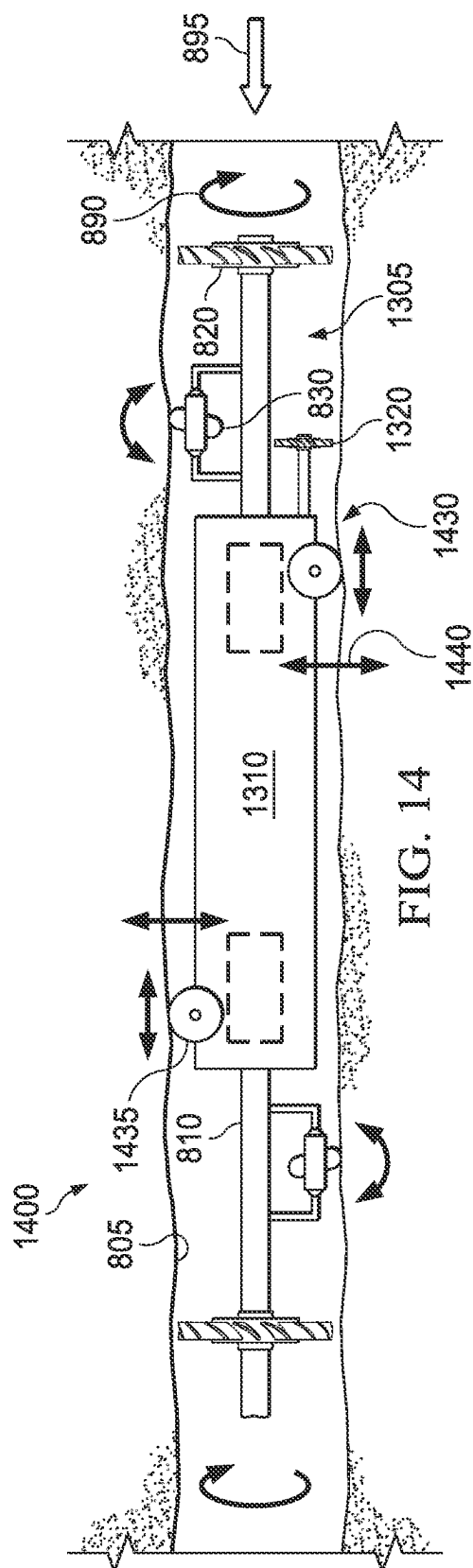


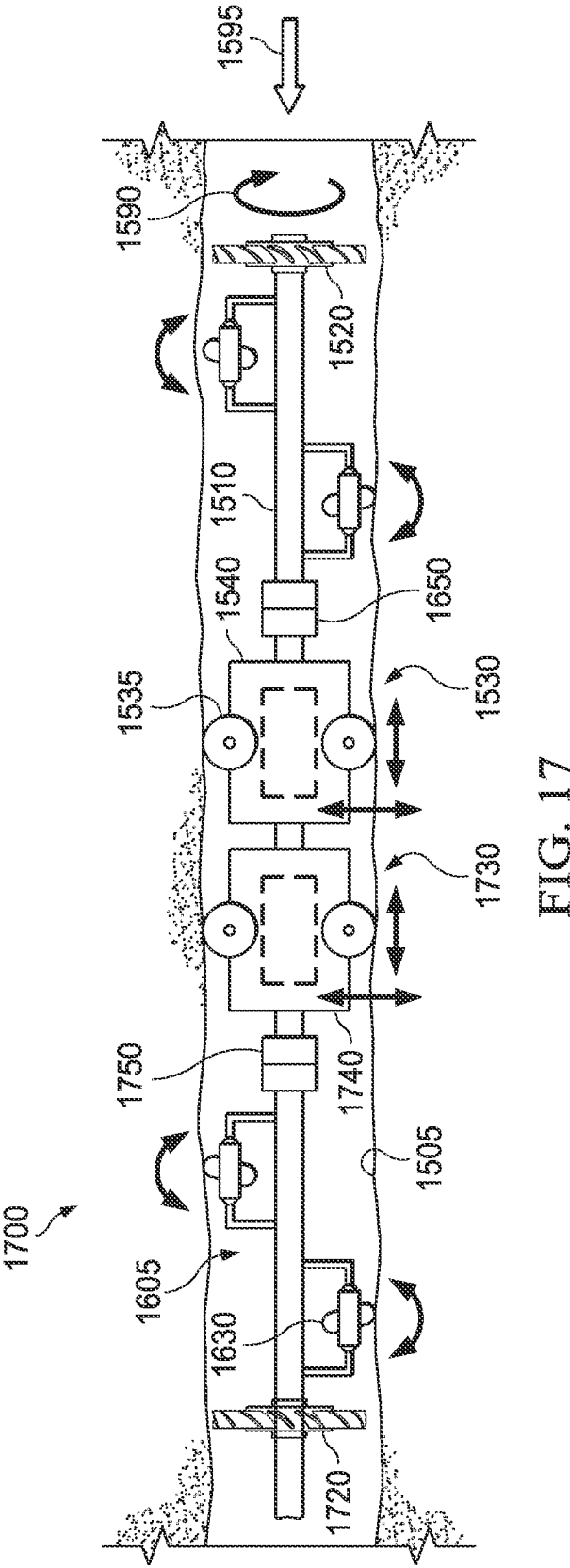
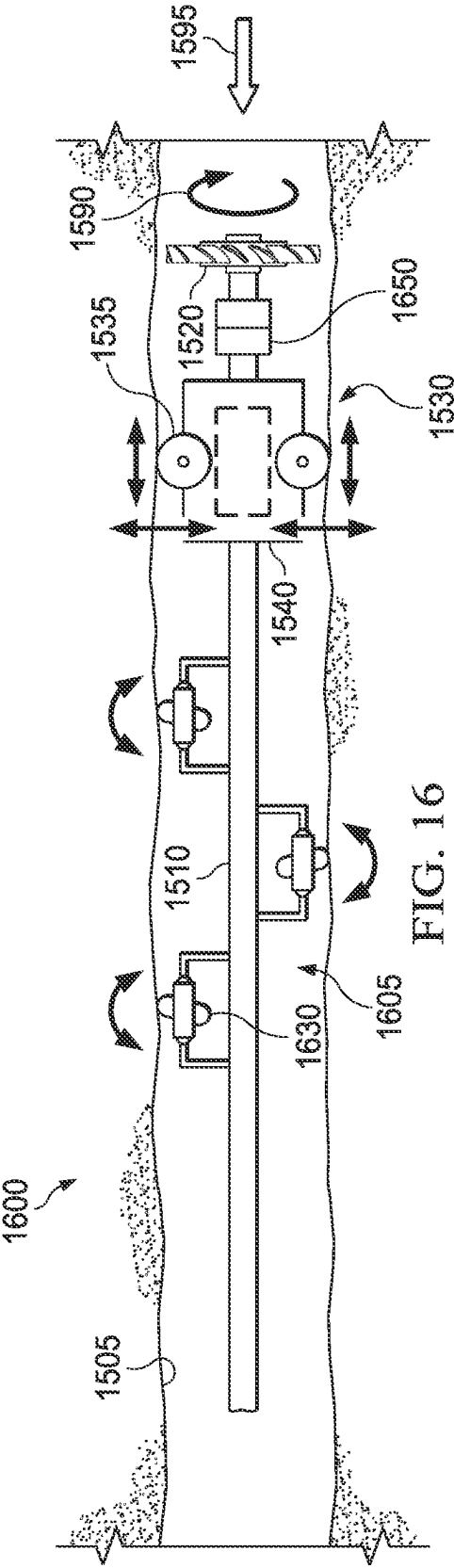
FIG. 7











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AUTONOMOUS TRACTOR USING COUNTER FLOW-DRIVEN PROPULSION

CROSS-REFERENCE TO RELATED APPLICATION

This application is a divisional of U.S. application Ser. No. 16/424,888, filed on May 29, 2019 entitled "AUTONOMOUS TRACTOR USING COUNTER FLOW-DRIVEN PROPULSION," which claims the benefit of U.S. Provisional Application Ser. No. 62/679,107, filed on Jun. 1, 2018 entitled "AUTONOMOUS TRACTOR USING COUNTER FLOW-DRIVEN PROPULSION," commonly assigned with this application and incorporated herein by reference in their entirety.

BACKGROUND

Intervention into the lateral is difficult and typically requires using coiled tubing, jointed tubing, or an e-line driven tractor. Coiled tubing is expensive, the injection lengths are limited by the size of the coil, and furthermore requires a coiled tubing rig to run. Jointed tubing is slow and also requires a rig to be moved into place.

As a result, many of the traditional wellbore interventions have used a tractor to pull a wireline tool. This traditional approach is limited by the weight of the wireline and the cost of the tractor. Autonomous downhole robotic tractors have been desired, but have been limited by battery weight and system cost. What is needed in the art is an improved autonomous downhole tractor that does not experience the drawbacks of existing systems.

BRIEF DESCRIPTION

Reference is now made to the following descriptions taken in conjunction with the accompanying drawings, in which:

FIGS. 1-7 illustrate various different embodiments of well systems manufactured, designed and operated according to the disclosure; and

FIGS. 8-17 illustrate various different embodiments of wellbore tractors manufactured, designed and operated according to the disclosure.

DETAILED DESCRIPTION

In the drawings and descriptions that follow, like parts are typically marked throughout the specification and drawings with the same reference numerals, respectively. The drawn figures are not necessarily, but may be, to scale. Certain features of the disclosure may be shown exaggerated in scale or in somewhat schematic form and some details of certain elements may not be shown in the interest of clarity and conciseness. The present disclosure may be implemented in embodiments of different forms. Specific embodiments are described in detail and are shown in the drawings, with the understanding that the present disclosure is to be considered an exemplification of the principles of the disclosure, and is not intended to limit the disclosure to that illustrated and described herein. It is to be fully recognized that the different teachings of the embodiments discussed herein may be employed separately or in any suitable combination to produce desired results. Moreover, all statements herein reciting principles and aspects of the disclosure, as well as specific examples thereof, are intended to encompass

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equivalents thereof. Additionally, the term, "or," as used herein, refers to a non-exclusive or, unless otherwise indicated.

Unless otherwise specified, use of the terms "connect," "engage," "couple," "attach," or any other like term describing an interaction between elements is not meant to limit the interaction to direct interaction between the elements and may also include indirect interaction between the elements described.

Unless otherwise specified, use of the terms "up," "upper," "upward," "uphole," "upstream," or other like terms shall be construed as generally toward the surface of the formation; likewise, use of the terms "down," "lower," "downward," "downhole," or other like terms shall be construed as generally toward the bottom, terminal end of a well, regardless of the wellbore orientation. Use of any one or more of the foregoing terms shall not be construed as denoting positions along a perfectly vertical or horizontal axis. Unless otherwise specified, use of the term "subterranean formation" shall be construed as encompassing both areas below exposed earth and areas below earth covered by water, such as ocean or fresh water.

The present disclosure teaches how to make a new class of wellbore tractor that operates without electricity. The wellbore tractor, in one embodiment, is mechanically powered by the flow of wellbore fluid. Thus, in one embodiment, neither wireline nor batteries are required to propel such a wellbore tractor. Notwithstanding, in certain embodiments batteries and/or wireline may be used in conjunction with the mechanical power to propel the wellbore tractor.

Such a wellbore tractor uses the wellbore fluid flow energy to propel the wellbore tractor forward. For example, in one embodiment the wellbore fluid flow energy spins the wellbore tractor such that it spirals into the flow. The result is a very low cost and very rugged wellbore tractor that enables a new class of applications for logging, communication, sealing, and wellbore evaluation.

The present disclosure further focuses on the aspect where the wellbore tractor acts as an autonomous robot. For instance, in such a use the wellbore tractor is placed in the well, the wellbore fluid is allowed to flow uphole, and as a result the wellbore tractor swims downhole to a specific location and performs a predefined job. The design of the wellbore tractor, in one embodiment allows the tool to travel downhole and uphole in vertical and horizontal sections of the wellbore.

A flow-powered wellbore tractor, according to the disclosure, uses the energy of the wellbore fluid (e.g., production fluid) to move forward. The flowing wellbore fluid, in one embodiment, hits the turbine on the front of the wellbore tractor, which causes the turbine to rotate in a first rotational direction, and thus the wellbore tractor to also rotate. In one embodiment, the wellbore tractor has one or more wellbore engaging devices attached thereto that are in contact with a surface of the wellbore. These wellbore engaging devices allow the wellbore tractor to rotate, as the wellbore engaging devices are (e.g., in one embodiment) tilted at a slight angle relative to an axial surface of the wellbore. This slight angle causes the wellbore tractor to advance a little bit into the wellbore with each rotation. Thus, the wellbore fluid flow causes the wellbore tractor to spiral upstream, somewhat like a drill.

A variety of different wellbore engaging devices may be used and remain within the purview of the disclosure. In one embodiment, the wellbore engaging devices are wheels. However, although the term "wheel" is used herein, the present disclosure contemplates that other rolling members,

such as tracks, roller bearings, or otherwise, may also be employed in lieu of or in addition to any illustrated wheels. In accordance with one embodiment, the wheels may comprise dissolvable wheels. Accordingly, downhole conditions, such as temperature, pressure, fluid type, etc. may be used to dissolve the wheels, and thus in certain embodiments allow the wellbore tractor to be pushed uphole by the wellbore fluid.

In one embodiment of the disclosure, the wellbore tractor could be made up of two sections: a drive section and an automation section. In accordance with this embodiment, the drive section would take the wellbore tractor downhole, for example using one or more of the ideas discussed above. The automation section, in contrast, would be used to perform a downhole task. For example, the automation section could be a logging tool, for logging information from the wellbore as the wellbore tractor moves downhole. In another embodiment, the automation section includes memory and a receiver for receiving information from one downhole device, includes memory and a transmitter for transmitting information to a downhole device, or memory and a transceiver for receiving information from one downhole device and transmitting information to another downhole device. In another embodiment, the automation section is a perforator tool, and thus may be used to perforate openings in the wellbore, wellbore casing, production tubing, etc. In yet another alternative embodiment, the automation section is a swellable packer tool that is configured to swell and thus deploy downhole.

In yet another embodiment, the automation section is a sleeve shifting tool having a profile configured to engage with a corresponding profile in a downhole sleeve, and thus may be used as a sleeve shifting tool. A contrast between the different methodologies of performing the work can be seen when comparing a downhole power unit (DPU) to shift a sleeve instead of jarring action controlled at the surface. A wellbore tractor according to the disclosure is more analogous to the DPU, where the wellbore tractor has a flow-powered drive section and an automation section for performing work. One focus of the disclosure is to use wellbore fluid flow to power the high energy demand wellbore tractor, and thus in one embodiment eliminate the slickline, e-line, tubing conveyed, or coiled tubing equipment generally required for a DPU.

Many different variations of the wellbore tractor are feasible, all of which are within the purview of the disclosure. In one variation, a wellbore tractor walks to or past a shifting sleeve, dogs latch into the profile, a chute or vanes are configured to catch the flow, and the well flow/pressure shifts the sleeve. In yet another variation, the wellbore tractor carries a lockout sleeve downhole. The lockout sleeve sets and one or more features of the wellbore tractor dissolves. For example, a lockout sleeve might block an inflow control device (ICD), and then the turbine of the wellbore tractor dissolves leaving an open passageway. In this example, the wellbore tractor is both the means of transportation and the tool. In yet another variation, the wellbore tractor travels downhole, engages a fishneck, blocks the flow by use of a chute or shifting vane, and then retrieves the tool having the fishneck.

In yet another variation, multiple different types of drive sections are used. For example, in addition to the flow based drive section discussed above, a powered drive section could also be used. For example, the flow based drive section could be used with one or both of a hydraulically powered drive section or an electrically powered drive section and remain within the scope of the disclosure. Accordingly to

one embodiment, the wellbore fluid flow drives the spinning motion as well as generates hydraulic and/or electric energy. This energy may be used for the drive section, for the automation section, or both.

In one example embodiment, the wellbore tractor (e.g., the base member coupled to the turbine) includes a slip clutch, where if the wellbore tractor body stops moving, the turbine continues to turn and flow is used to generate hydraulic and/or electric power that is used for propulsion. An example would be that the wellbore tractor moves downward rapidly and reaches a difficult spot causing the wellbore tractor to stop. The turbine continues to spin and a secondary propulsion system that is stronger but slower kicks in to get it over the difficult part. Once past the difficult part, the primary propulsion system may resume. In one embodiment, a first turbine could provide power for the drive section, and a second turbine could provide power to the automation section (e.g., a sensor or another device). Thus, a wellbore tractor according to the disclosure could have multiple turbines and pumps/generators that do different things.

In one variation, the drag in the pumps/generators impart the turbine momentum to the entire wellbore tractor. If the torque required to spin the wellbore tractor is higher than the drag, the turbine and the tractor body turn at different speeds and the speed difference generates hydraulic/electric power. This also acts as a governor allowing the turbine to turn faster with higher flow rates and limiting the rotational speed of the body. In contrast, a fixed turbine drag increases proportionally faster with increased flow rates. In yet another variation, the body of the wellbore tractor does not spin, just the turbine. The turbine powers a hydraulic pump or drives an electrical generator that is used to drive hydraulic or electrical motors. In this variation, multiple energy harvesting turbines may be used, and furthermore there may be multiple systems that power the propulsion system. Furthermore to this variation, multiple systems can result in a specially tailored function. For example the electric motors provide a fast but weak mode of propulsion, but if the wellbore tractor reaches a "tough" spot in the well, the hydraulic system, which is slow but strong, propels the tractor past the tough spot.

In yet another variation, a means of controlling the device can be to close in the well. When the well closes in, pressure builds which may trigger an atmospheric chamber to shear pins and provide work. An example would be a wellbore tractor that finds a profile. When the surface valve is closed, pressure builds and the atmospheric chamber is triggered, and then slips are deployed and the downhole sleeve is shifted.

An additional variation includes when the wellbore tractor is used to transport a perforator tool to a specific location. Manipulation of the surface valve sends a signal to the guns causing them to fire. The guns are either left in the well, dissolve, or are retrieved by the wellbore tractor. In yet another variation, low flow results in the wellbore tractor swimming against the flow and high flow cause the wellbore tractor to flow out of the well. In this application, one could control the flow rate (e.g., from the surface of the well) to reverse the direction of the axial movement of the wellbore tractor. For example, with a slow flow rate the friction between the wellbore tractor and the wellbore is sufficient to keep the tractor moving in the correct direction, but once a higher flow rate is encountered, the friction between the wellbore tractor and the wellbore is not sufficient to keep the wellbore tractor moving in the correct direction, and thus the

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wellbore tractor will now move in the opposite direction with the high flow of the fluid.

In yet another variation, the wellbore tractor carries a battery downhole and leaves it there to power existing equipment, or the wellbore tractor places equipment in the well, such as a frac plug. In yet another variation, a battery is used to provide supplemental propulsion to the wellbore tractor. The wellbore tractor relies upon well flow for primary propulsion, but facing a high power demand action, the supplemental energy from the battery may be used to supplement the power needs. In another variation, the well-

bore tractor is covered with a swellable packer or carries a swellable packer, and thus can operate as a bridge plug. This disclosure further focuses on the employment of a simple device that has limited function. The idea is for a simple device that can travel in the counter flow direction and trigger a mechanism. This leads to the possibility of extending the technology of dart/ball into the horizontal section of the completion. Furthermore a means of retrieving the wellbore tractor may be built into the mechanism for its return to the surface. Using a dropped dart/ball to initiate a downhole action is known in the industry. However, dropping balls is limited by gravity. In horizontal sections, the dart/ball must be pumped down into the horizontal sections. Pumping a dart/ball into the well uses a lot of water and has the potential to damage the formation.

This disclosure also includes a dart/ball that can "swim" upstream through vertical and horizontal sections of the well. The ball uses a mechanical flow-driven wellbore tractor to move upstream into the production flow. The energy of the production flow is used to mechanically power the ball. Basically this is a device that swims against the flow.

This disclosure also focuses on the aspect where the wellbore tractor is used to transport an untethered object to trigger a downhole action. The untethered object can be a dart, a ball, a frac plug, a baffle, a bridge plug, a wiper plug, or any other downhole tool. The wellbore tractor is placed in the well, the wellbore fluid is allowed to flow, and the wellbore tractor spirals downhole (including horizontal sections). In one embodiment, the wellbore tractor transverses downhole and triggers a downhole tool. After the downhole tool is triggered, pumping into the well may provide the force to perform work. Once the downhole tool is triggered, the wellbore tractor may return to the surface, may dissolve downhole, or may simply stay in the wellbore.

A wellbore tractor according to the disclosure functions in much the same way as a dropped dart/ball except that it is not gravity driven. The wellbore tractor swims to a location near a seat/receptacle/trigger. Typically the wellbore tractor swims past/to a trigger and activates the downhole tool. For example a spring loaded flapper is propped open and the wellbore tractor causes it to release and close. With the flapper closed, pressure from uphole/downhole provides force to perform work and manipulate the well. This pressure could further cause the flapper to reopen, and thus the wellbore tractor could continue its travel in the counter flow direction.

In one variation, wellbore fluid flow causes the wellbore tractor to travel downhole. At the furthest distance of its travel a mechanism on the wellbore tractor deploys/closes and instantly the wellbore tractor has a much greater flow restriction. The wellbore fluid flow causes the wellbore tractor to move upward, wherein the wellbore tractor lands in the down most receptacle. The well pressure then builds and shifts a sleeve. As the wellbore tractor is self-releasing, it may then move to the next receptacle where the process is repeated.

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In another variation, similar to that discussed in the paragraph above, the wellbore tractor travels downward with low flow and upward with high flow. Accordingly, manipulation at the surface could control the direction of movement of the wellbore tractor. In this embodiment, the wellbore tractor might land in a receptacle, wherein the receptacle triggers a change in the wellbore tractor. The wellbore tractor then seals off in the receptacle, and uphole/downhole pressure is utilized to shift the receptacle. In another variation, the pressure is used to set a packer being delivered downhole with the wellbore tractor. In another variation, a counting mechanism could be built into the wellbore tractor, and thus the counting mechanism could for example cause the device to set in a specific receptacle (e.g., the third receptacle).

The ability to log in horizontal sections is limited to tractor driven tools, coiled tubing logging, pump down tools, or tubing conveyed methods. Those interventions require some sort of rig, are limited in distance for evaluating the horizontal, and are expensive. Self-powered devices have previously been limited to battery-powered electrical motors. This disclosure additionally discloses how to build a self-powered wellbore tractor and focuses on the aspect where the wellbore tractor is used to log a wellbore. In one example, a logging instrument may be mechanically affixed to the tractor (or attached to the device). The wellbore tractor may then be placed in the well, and the wellbore fluid is allowed to flow. The wellbore tractor may then spiral its way downhole and through the horizontal sections, logging the well as it goes. It can then get produced out of the hole, for example using a chute, or in another embodiment, the wellbore fluid alone. In another embodiment, the wellbore tractor traverses to the bottom of the well, and thus logs the formation as it travels back up to the surface.

Such a wellbore tractor may be used to provide a low-cost intervention to a wellbore. In one embodiment, the wellbore tractor is used to provide simple logging in a low-cost wellbore. The production flow causes the wellbore tractor to spiral into the wellbore. The wellbore tractor is carrying sensor electronics for logging the wellbore. The sensor electronics could include a power source (battery or turbine generator) as well as either memory and/or a wireless transmitter. The wellbore tractor is logging as it spirals upstream. After the wellbore tractor has completed its mission, the memory is allowed to return to the surface. In one example, one or all of the wheels or the turbine dissolve and the sensor electronic package is produced back to the surface. In other examples, the entire tractor could dissolve and only the memory is released.

While a simple wellbore tractor would provide the function needed, it could be designed with a number of additional features/variants. For example, all or part of the wellbore tractor could dissolve, allowing the instrument package to return to the surface by wellbore flow. Alternatively, at the bottom of the well, the wellbore tractor could deploy a "chute" that results in wellbore fluid flow returning the wellbore tractor to the surface. In another variation, the wellbore tractor is partially composed of a syntactic foam which reduces the density of the wellbore tractor and more easily allows it to be produced to the surface.

In an alternative variation, vanes on the wellbore tractor are reversed and the wellbore tractor walks out of the well powered by well flow. This reversal could be initiated by: increased flow, temperature, pressure, time, electronics, dissolution of a catch, etc. In another variation, the tilt of the wheels is reversed and the wellbore tractor walks out of the well powered by well flow. This reversal could be initiated

by: increased flow, temperature, pressure, time, electronics, dissolution of a catch, etc. In an alternative embodiment, the wellbore tractor includes a first section used to travel downhole and a second section used to travel uphole. In this embodiment, one of the sections might be disabled when the other is active.

In an alternative embodiment, a logging device coupled to the wellbore tractor transmits a signal as it logs and thus retrieval is optional. For example, the logging device could create an acoustic signal that is received by a distributed acoustic sensing (DAS) fiber optic cable or an acoustic signal that is received by an acoustic transceiver (e.g., DynaLink wireless telemetry). The logging device could create an electromagnetic (EM) signal that is received by a EM transducer.

The wellbore tractor can be designed to work in open hole, cased hole, or completion tubing. Moreover, the wellbore tractor could be designed so that gravity is allowed to propel the wellbore tractor downhole in vertical sections. For example flow of wellbore fluid is stopped in the well when the wellbore tractor is inserted, the wellbore tractor “falls” to the horizontal section, then the flow of wellbore fluid is increased to propel the wellbore tool further downhole through the horizontal section.

While the array of logging applications performed with a wellbore tractor according to the disclosure is only limited by the logging instrumentation, there are certain applications that are very well suited for this type of wellbore tractor. For example, such a wellbore tractor may be used to survey along the length of the wellbore for temperature, flow composition, flow rate, flow noise, or pressure, or may be used to survey valve position, component health, wellbore health, scale formation, corrosion, leaks, etc. In another application, the wellbore tractor goes downhole and act as seismic sensor for a thumper being driven at the surface, or in an alternative embodiment the wellbore tractor “pings” and the signal is interpreted at the surface.

In another application, the wellbore tractor takes one or more samples at various depths. These samples are recovered when the wellbore tractor is retrieved. A simple methodology would be for simple vacuum chambers to be fitted with a rupture disk and a onetime check valve. At the prescribed pressure, the disk ruptures and a sample is taken. The onetime check valve prevents fluid from entering and leaving the chamber after the initial sample is taken. Another methodology would be for the vacuum chamber inlets to be controlled by time. Yet another methodology would be for the vacuum chamber inlet to be controlled by temperature.

Additionally, intervention-less logging could be achieved in a subsea well. A remotely operated vehicle (ROV) could transfer the wellbore tractor to a lubricator. The lubricator would open the path to the well and gravity would place the wellbore tractor in the well. A “snatch” mechanism could be built into the lubricator to pull the wellbore tractor the last few feet into the lubricator.

The ability to communicate with downhole equipment and downhole sensors is difficult. Typically communication is accomplished with an expensive wired cable or with power-intensive wireless communication. Reducing the distance for wireless communication will reduce the power consumption and can open new technologies for wireless data transfer. Reducing the wireless communication distance has been achieved by moving the transmitter and receiver closer to each other by lowering an acoustic transceiver on wireline, but this approach does not work in horizontal sections. Reducing the wireless transmission distance in horizontal sections requires the use of tractor driven, coiled

tubing, pump down, or tubing conveyed methods which are all expensive and require a rig.

This disclosure also embodies the idea of achieving wireless communication by using a wellbore tractor that is mechanically powered by the production flow. This is a new class of wellbore tractor that propels itself without electricity. Basically this is a wellbore tractor that swims against the flow in order to relay commands, data, and information.

This disclosure also focuses on the aspect where the wellbore tractor acts as a messenger to send or receive information in a well. In one example, the wellbore tractor is placed in the well, the wellbore fluid is allowed to flow, and thus the wellbore tractor spirals downhole (including horizontal sections). As the wellbore tractor passes other pieces of equipment information is broadcasted and/or received. Once the message is transferred, the wellbore tractor may (or may not) return to the surface.

For example, a wellbore tractor according to the disclosure may be used to provide a low-cost wireless communication in a wellbore. In one embodiment, the wellbore tractor is used to carry data between a downhole location and the surface. The production flow causes the wellbore tractor to spiral into the wellbore. The wellbore tractor, in this embodiment, is carrying transceiver electronics for communicating with downhole tools in the wellbore. The transceiver electronics could include a power source (battery or turbine generator), a wireless transceiver, and support electronics. In one example, the wellbore tractor spirals upstream past the downhole tools, and as it passes the downhole tools it relays data with the tool. In one example, the transceiver electronics receives sensor data from a downhole flow sensor and transmits a new position command to an inflow control valve (ICV). After the wellbore tractor has completed its mission, the memory in the electronics is allowed to return to the surface so that the operator can receive the sensor data. In one example, the wheels and the turbine blades dissolve and the sensor electronic package is produced back to the surface. In other examples, the entire tractor could dissolve and only the memory is released.

In one application, the wellbore tractor carries the data entirely back to the surface. Alternatively, the wellbore tractor could carry the data back to a transmission hub where the hub sends the data back to the surface. The transmission hub could be a DynaLink-style acoustic transmitter. The transmission hub could be a wired connection on an upper completion and the wellbore tractor is relaying data from the unwired lower completion to the wired upper completion. Finally, the transmission hub could be in the main bore and the wellbore tractor is carrying information out of a lateral.

Other applications also exist. In one example, the wellbore tractor carries information downhole and transfers it at the appropriate location. For example the wellbore tractor tells the ICV to change setting, for example based upon instruction predetermined at the surface. In another embodiment, the wellbore tractor includes sensors that sense the well, and the wellbore tractor uses the information gained from traveling downhole, to tell the ICV or eICD to readjust.

In an alternative embodiment, the wellbore tractor travels downhole and records RFID information as it logs. The wellbore tractor signals the RFID and receives information back. The downhole equipment can be completely passive with both the signal and receiving function contained within the wellbore tractor. The information for example can be like “ICD-4 is 25% open”. This can also be accomplished with other magnetic, electrical, or electromagnetic transmission such as near field communication or radio signals. The short-hop wireless signal could also be acoustic or vibration

based transmission. In another embodiment, the wellbore tractor uses RFID information as it logs, and processes that information so that it can tell other equipment what to do. For example "Since ICD-4 is 25% open then close ICD-5 an additional 5%". In this embodiment, the wellbore tractor functions as a power source (e.g., broadcaster). For example, the wellbore tractor travels downhole transmitting a power source. While the downhole tools are passive, when the wellbore tractor is proximate thereto the downhole tools become active.

The application of wireline conveyed tools are limited by friction of the wirelines and need a tractor to enter a horizontal wellbore. While electrically-driven tractors have been developed for electrical wireline, equivalent tractors have not been developed for slickline or sandline. Theoretically, a battery-powered tractor could be developed but the operational life would be limited and the pulling power of a battery-powered tractor would be limited. Tractors for electrical wireline are expensive, heavy, and can only be placed at the end of the wireline. Thus, there is a need for a mechanically-driven wellbore tractor for wireline applications. There is also a need to be able to place these wellbore tractors not only at the end of the wireline but also at intermediate locations to help carry the weight of the wire.

One aspect of this idea is that some of the production flow energy can be used as propulsion for the wireline. One idea is for a wireline wellbore tractor to use wellbore flow to produce some if not all of its propulsion energy demand. The wellbore tractor can be used on the end of the wire, and furthermore a second wellbore tractor can be used as a clamp-on configuration to help support an intermediate location of the wireline and to help reduce the tension on the wire. Such wellbore tractors are likely to be very inexpensive, and in one embodiment the wellbore tractor is used to carry fiber optic cable into the internal diameter (ID) of the tubing. The fiber optic cable can use DTS and DAS to provide real-time understanding of the production, and deployment on the ID allows for installation after the well is operational and may be simpler and less expensive. For fiber optic deployment, the wellbore tractor could be considered a disposable item. In one application, the wellbore tractor dissolves downhole. In another application, the wellbore tractor serves as an anchor for the fiber optic cable.

One goal is for the wellbore tractor to drag the wire and slickline tools to the desired location in the well and through the horizontal section. After work is done, the slickline rig can retrieve the tool string and tractor. Means to accomplish this task and variations include, without limitation: A) The wellbore tractor runs past a sleeve and the wireline tool shift the sleeve when the device is retrieved. Specially designed spring loaded detent jars may be required to jar in the horizontal; B) A DPU device is attached to the slickline string to perform the desired work; C) The wellbore tractor shifts into reverse and helps retrieve the tool/wire; D) Multiple wellbore tractors pull the wire. For example, wellbore tractor devices could be added to the wire as it is unspooled into the well. The wellbore tractors may reverse, helping to retrieve the wire. For example a wellbore tractor may be deployed every 1000 ft., among other locations. Optionally, a signal at the lubricator could signal the wellbore tractor to attach/detach from the wire.

Referring to FIG. 1, depicted is a well system 100 including an exemplary operating environment that the apparatuses, systems and methods disclosed herein may be employed. For example, the well system 100 could use a wellbore tractor according to any of the embodiments, aspects, applications, variations, designs, etc. disclosed in

the preceding and/or following paragraphs. The illustrated well system 100 initially includes a wellbore 110. The illustrated wellbore 110 is a deviated wellbore that is formed to extend from a terranean surface 120 to a subterranean zone 130 (e.g., a hydrocarbon bearing geologic formation) and includes a vertical portion 140, a radius portion 145, and a horizontal portion 150. Although portions 140 and 150 are referred to as "vertical" and "horizontal," respectively, it should be appreciated that such wellbore portions may not be exactly vertical or horizontal, but instead may be substantially vertical or horizontal to account for drilling operations. Further, the wellbore 110 may be a cased well, a working string or an open hole, and is of such length that it is shown broken.

Further, while the well system 100 depicted in FIG. 1 is shown penetrating the earth's surface on dry land, it should be understood that one or more of the apparatuses, systems and methods illustrated herein may alternatively be employed in other operational environments, such as within an offshore wellbore operational environment for example, a wellbore penetrating subterranean formation beneath a body of water.

In the illustrated embodiment of FIG. 1, a wellbore tractor 160 manufactured and designed according to the disclosure is positioned within a wellbore 110. In accordance with one embodiment of the disclosure, the wellbore tractor 160 is mechanically powered by wellbore fluid 190. In this instance, energy from the wellbore fluid 190 spins the wellbore tractor 160, causing the wellbore tractor 160 to spiral into the flow.

The wellbore tractor 160 illustrated in FIG. 1 includes a base member 165. The base member 165 is illustrated as a shaft in the illustrative embodiment, but may comprise many different designs and/or sizes and remain within the scope of the disclosure. A single turbine 170 is fixed to the base member 165 in the embodiment of FIG. 1. The term "fixed" as used herein, means that the base member 165 and the one or more turbines 170 rotate as a single unit. The term "turbine" as used herein, is meant to include a structure having two or more blades or vanes that are positioned to induce rotation. Given the foregoing, the turbine 170 is configured to rotate the base member 165 in a first rotational direction 195 based upon a first direction of fluid flow, which in the illustrated embodiment is the fluid 190. It should be noted that while a single turbine 170 is fixed to the base member 165 in the illustrated embodiment of FIG. 1, other embodiments exist wherein additional turbines are coupled and/or fixed to the base member 165, as will be further discussed below.

The wellbore tractor 160 additionally includes one or more wellbore engaging devices 175 radially extending from the base member 165. In accordance with one embodiment of the disclosure, the one or more wellbore engaging devices 175 are contactable, and in fact in contact with, a surface of the wellbore 110. Accordingly, the one or more wellbore engaging devices 175 displace the base member 165 and turbine 170 axially downhole as the turbine 170 rotates in the first rotational direction 195, for example in response to the flow of the fluid 190 there past. The wellbore tractor 160 illustrated in FIG. 1 is fairly simple in design, and thus does not include one or more of the other aspects, including an automation section, for example as discussed above.

In accordance with the disclosure, the fluid 190, which is production fluid in one embodiment, may be controlled from the surface of the wellbore 110. For example, in one embodiment, the velocity of the flow of production fluid 190 may

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be controlled from the surface of the wellbore **110** to speed up or slow down the displacement of the wellbore tractor **160** axially downhole. In another embodiment, the velocity of the flow of production fluid **190** may be increased (e.g., from the surface) to a value sufficient to overcome friction between the one or more wellbore engaging devices **175** and the surface of the wellbore **110** and thus push the wellbore tractor **160** uphole. In yet another embodiment, the wellbore tractor **160** may be subjected to fluid (e.g., fluid from the surface) in a second opposite direction to rotate the base member **165** in a second opposite rotational direction to displace the wellbore tractor **160** axially uphole.

Turning to FIG. 2, illustrated is a well system **200** having an alternative embodiment of a wellbore tractor **260** manufactured and designed according to the disclosure. The well system **200** and wellbore tractor **260** share many elements with the well system **100** and wellbore tractor **160** illustrated in FIG. 1. Accordingly, like reference numerals may be used to indicate similar, if not identical, features. In the embodiment of FIG. 2, the base member **165**, turbine **170**, and one or more wellbore engaging devices **175** form at least a portion of a flow based drive section, and the wellbore tractor **260** additionally including an automation section **270** for performing a downhole task. In the particular embodiment of FIG. 2, the automation section **270** is a logging tool **280**. The logging tool **280**, in one embodiment, includes a power source **282**, memory **284**, and a wireless transmitter **286**, among other relevant features, and is configured to log one or more parameters of the wellbore **110**. In one embodiment, the logging tool **280** logs the wellbore **110** as it is displaced axially downhole. In another embodiment, the logging tool **280** travels to a downhole end of the wellbore **110**, and then logs the wellbore **110** as it travels axially uphole. As discussed above, the logging tool **280**, or at least the memory **284** thereof, may return uphole using the flow of the fluid **190**, or alternatively using a chute **290**.

Turning to FIG. 3, illustrated is a well system **300** having an alternative embodiment of a wellbore tractor **360** manufactured and designed according to the disclosure. The well system **300** and wellbore tractor **360** share many elements with the well system **200** and wellbore tractor **260** illustrated in FIG. 2. Accordingly, like reference numerals may be used to indicate similar, if not identical, features. In the particular embodiment of FIG. 3, the wellbore tractor **360** includes an automation section **370** including memory **382** and a transceiver **386**. The memory **382** and transceiver **386**, in the embodiment of FIG. 3, are configured to receive information from one downhole device **388** (e.g., a flow sensor in one embodiment) and transmit the information to another downhole device **390** (e.g., an ICV in one embodiment), for example wirelessly, as discussed above. While it has been illustrated in FIG. 3 that the downhole device **388** transmits the information and the downhole device **390** receives the information, the opposite may also be true.

Turning to FIG. 4, illustrated is a well system **400** having an alternative embodiment of a wellbore tractor **460** manufactured and designed according to the disclosure. The well system **400** and wellbore tractor **460** share many elements with the well system **200** and wellbore tractor **260** illustrated in FIG. 2. Accordingly, like reference numerals may be used to indicate similar, if not identical, features. In the particular embodiment of FIG. 4, the wellbore tractor **460** includes an automation section **470**. The automation section **470**, in the embodiment of FIG. 4, is a perforator tool **480**. The perforator tool **480**, as those skilled in the art appreciate, may be

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taken downhole using the wellbore tractor **460** and then used to perforate the wellbore **110**, as discussed in greater detail above.

Turning to FIG. 5, illustrated is a well system **500** having an alternative embodiment of a wellbore tractor **560** manufactured and designed according to the disclosure. The well system **500** and wellbore tractor **560** share many elements with the well system **200** and wellbore tractor **260** illustrated in FIG. 2. Accordingly, like reference numerals may be used to indicate similar, if not identical, features. In the particular embodiment of FIG. 5, the wellbore tractor **560** includes an automation section **570**. The automation section **570**, in the embodiment of FIG. 5, is a sleeve shifting tool **580**. The sleeve shifting tool **580**, in one embodiment, has a profile **582** configured to engage with a corresponding profile **592** in a downhole sleeve **590**. The sleeve shifting tool **580**, as those skilled in the art appreciate, may be taken downhole using the wellbore tractor **560** and then used to shift the downhole sleeve **590**, as discussed in greater detail above.

Turning to FIG. 6, illustrated is a well system **600** having an alternative embodiment of a wellbore tractor **660** manufactured and designed according to the disclosure. The well system **600** and wellbore tractor **660** share many elements with the well system **200** and wellbore tractor **260** illustrated in FIG. 2. Accordingly, like reference numerals may be used to indicate similar, if not identical, features. In the particular embodiment of FIG. 6, the wellbore tractor **660** includes an automation section **670**. The automation section **670**, in the embodiment of FIG. 6, is a swellable packer tool **680**. The swellable packer tool **680**, as those skilled in the art appreciate, may be taken downhole using the wellbore tractor **660** and then swell to function as a packer or isolation plug, as discussed in greater detail above.

Turning to FIG. 7, illustrated is a well system **700** having an alternative embodiment of a wellbore tractor **760** manufactured and designed according to the disclosure. The well system **700** and wellbore tractor **760** share many elements with the well system **100** and wellbore tractor **160** illustrated in FIG. 1. Accordingly, like reference numerals may be used to indicate similar, if not identical, features. In the particular embodiment of FIG. 7, the wellbore tractor **760** is coupled proximate a downhole end of a wireline **770**. The term wireline, as used in this embodiment, is intended to include traditional wireline, slickline, sandline, e-line, braided cable, fiber-optic cable, etc. In the embodiment of FIG. 7, the wireline **770** may be pulled downhole using the wellbore tractor **760**. Further to the embodiment of FIG. 7, the wellbore tractor **760** is a first wellbore tractor, and the well system **700** further includes a second wellbore tractor **780**. The second wellbore tractor **780**, as shown, may be coupled to an intermediate location of the wireline **770** uphole of the first wellbore tractor **760**. In accordance with one embodiment of the disclosure, the first and second wellbore tractors **760**, **780** may also help return the wireline **770** uphole.

Turning now to FIGS. 8 to 17, illustrated are different embodiments of wellbore tractors manufactured and designed according to the disclosure. The wellbore tractors illustrated in FIGS. 8 to 17 differ from one another primarily in the way their drive section operates. As the wellbore tractors illustrated in FIGS. 8 to 17 share many features, similar reference numerals may be used to indicate similar, if not identical, features. Additional details for each of the wellbore tractors illustrated in FIGS. 8 to 17 may be found in the preceding paragraphs.

FIG. 8 illustrates one simple embodiment of a wellbore tractor **800** manufactured and designed according to the disclosure, and placed within a wellbore **805**. The wellbore

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tractor **800** includes a base member **810**, and a turbine **820** fixed to the base member **810**. As is well understood by now, the turbine **820** is designed to rotate the base member **810** in a first rotational direction **890** based upon a first direction of fluid flow, such as may occur with the fluid **895**. The wellbore tractor **800** additionally includes a plurality of wellbore engaging devices **830** radially extending from the base member **810**. The plurality of wellbore engaging devices **830**, in the illustrated embodiment, are contactable with a surface of a wellbore **805** for displacing the base member **810** and turbine **820** axially downhole as the turbine **820** rotates in the first rotational direction **890**.

In accordance with the embodiment of FIG. **8**, the one or more wellbore engaging devices **830** are one or more wheels. For example, the wheels are positioned at a first tilted direction relative to an axial surface of the wellbore **805**. Accordingly, the tilted wheels are configured to displace the base member **810** and turbine **820** axially downhole.

In accordance with one embodiment of the disclosure, the one or more wellbore engaging devices **830**, or other features of the wellbore tractor **800**, are dissolvable in response to a downhole condition. For example, the one or more wellbore engaging devices **830**, or the other features of the wellbore tractor **800** including the turbine **820**, may be dissolvable in response to time, temperature, pressure, or fluid type, among other downhole conditions. As discussed in greater detail above, the dissolvable nature of the wellbore tractor **800** allows different parts, or the entirety, of the wellbore tractor **800** to return to the surface of the wellbore **805**, in certain instances simply using the flow of the fluid **895**.

Turning to FIG. **9**, illustrated is an alternative embodiment of a wellbore tractor **900** manufactured and designed according to the disclosure. The wellbore tractor **900** primarily differs from the wellbore tractor **800** of FIG. **8** in that the wellbore tractor **900** includes a second turbine **920a**, and in this particular embodiment a third turbine **920b**. In the illustrated embodiment, the second turbine **920a** and turbine **820** are fixed at opposing ends of the base member **810**. Additionally, the third turbine **920b** is positioned between the second turbine **920a** and the turbine **820**, for example substantially at a midpoint between the two. The second and third turbines **920a**, **920b**, in the illustrated embodiment, have the same orientation or handedness as the turbine **820**, and thus are also configured to rotate the base member **810** in the first rotational direction **890**. In the illustrated embodiment of FIG. **9**, the turbines **820**, **920a**, **920b** are configured to rotate the base member clockwise to advance the wellbore tractor **900** downhole (e.g., as looking up at the wellbore tractor **900** from downhole). The second and third turbines **920a**, **920b**, as those skilled in the art now understand, provide additional torque for displacing the wellbore tractor **900** axially downhole.

Turning to FIG. **10**, illustrated is an alternative embodiment of a wellbore tractor **1000** manufactured and designed according to the disclosure. The wellbore tractor **1000** primarily differs from the wellbore tractor **800** of FIG. **8** in that the one or more wellbore engaging devices **830** are one or more wheels **1030**, and the wellbore tractor **1000** further includes one or more wheel actuation members **1035** coupled to the one or more wheels **1030**. In the illustrated embodiment of FIG. **10**, the one or more wheel actuation members **1035** are configured to adjust an angle of tilt of the one or more wheels **1030** relative to the axial surface of the wellbore **805**. Such adjustments to the angle of tilt may be used for speeding up or slowing down the displacement of

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the wellbore tractor **1000** axially downhole. The one or more wheel actuation members **1035**, in certain embodiments, may also be used to move the one or more wheels **1030** from the first tilted direction (e.g., as illustrated in FIG. **10**) to a second opposite tilted direction (not shown) relative to the axial surface of the wellbore **805** for displacing the wellbore tractor **1000** axially uphole.

The wellbore tractor **1000** illustrated in FIG. **10**, in one embodiment, may further include one or more turbine actuation members **1040** coupled to the one or more turbines. In the illustrated embodiment of FIG. **10**, the one or more turbine actuation members **1040** are configured to adjust an angle of tilt of the one or more turbines, and more particularly their blades and/or vanes, relative to the first direction of fluid flow. Accordingly, the one or more turbine actuation members **1040** may be used to speed up or slow down the displacement of the wellbore tractor **1000** axially downhole, as well as potentially reverse totally, wherein the wellbore tractor **1000** can be displaced axially uphole using the same direction of fluid flow.

The one or more wheel actuation members **1035** may be operable in response to a variety of different signals or conditions. In one embodiment, the one or more wheel actuation members **1035** receive a signal from uphole. In another embodiment, the one or more wheel actuation members **1035** are operable in response to a downhole signal generated in the wellbore tractor **1000**. For example, the one or more wheel actuation members **1035** could move in response to changes in time, temperature or pressure, among other conditions measured in the wellbore tractor **1000**, particularly when changing from a state that moves the wellbore tractor **1000** axially downhole to an opposite state that moves the wellbore tractor **1000** axially uphole. The one or more turbine actuation members **1040** may also be operable in response to a variety of different signals or conditions, including the same signals or conditions as the one or more wheel actuation members **1035**.

Turning to FIG. **11**, illustrated is an alternative embodiment of a wellbore tractor **1100** manufactured and designed according to the disclosure. The wellbore tractor **1100** differs significantly from the wellbore tractor **800** of FIG. **8**. In the embodiment illustrated in FIG. **11**, the base member **810** is a first base member **1110a**, the one or more turbines **820** are one or more first turbines **1120a**, and the one or more wellbore engaging devices **830** are one or more first wellbore engaging devices **1130a**. The wellbore tractor **1100** illustrated in FIG. **11** further includes a second base member **1110b**, and one or more second turbines **1120b** fixed to the second base member **1110b**. The one or more second turbines **1120b**, in the illustrated embodiment, are operable for rotating the second base member **1110b** in a second opposite rotational direction **1190** based upon the first direction of fluid flow **895**. Accordingly, the one or more second turbines **1120b** have an opposite orientation or handedness as the one or more first turbines **1120a**. For example, while the one or more first turbines **1120a**, in the illustrated embodiment, are configured to rotate the first base member **1110a** clockwise to advance the wellbore tractor **1100** downhole (e.g., as looking up at the wellbore tractor **1100** from downhole), the one or more second turbines **1120b**, in the illustrated embodiment, are configured to rotate the second base member **1110b** counter clockwise to advance the wellbore tractor **1100** downhole (e.g., as looking up at the wellbore tractor **1100** from downhole).

The wellbore tractor **1100** additionally includes one or more second wellbore engaging devices **1130b** radially extending from the second base member **1110b**. In this

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embodiment, the one or more second wellbore engaging devices **1130b** are also contactable with the surface of the wellbore **805** for displacing the second base member **1110b** and one or more second turbines **1120b** axially downhole as the one or more second turbines **1120b** rotate in the second opposite rotational direction **1190**. In the embodiment of FIG. **11**, the first and second base members **1110a**, **1110b** are rotatably coupled to one another to allow for the first rotational direction **890** and second opposite rotational direction **1190**. For example, a swivel **1150**, or another similar device, could be used to rotatably couple the first and second base members **1110a**, **1110b** to one another.

Turning to FIG. **12**, illustrated is yet another embodiment of a wellbore tractor **1200** manufactured and designed according to the disclosure. The wellbore tractor **1200**, in addition to many of the features of wellbore tractor **800**, additionally includes a stator member **1210** rotatably surrounding at least a portion of the base member **810**. In the embodiment of FIG. **12**, the stator member **1210** has one or more stator turbines **1220** (e.g., a single turbine in this embodiment) fixed thereto for reversing a swirling action of fluid flow exiting the one or more turbines **820**. In one embodiment of the disclosure, such as shown in FIG. **12**, blades of the one or more stator turbines **1220** have an opposite orientation or handedness to blades of the one or more turbines **820**. Such a configuration helps to reverse the swirling action, and thus provide more torque for the wellbore tractor **1200**.

The wellbore tractor **1200** additionally includes one or more stator wellbore engaging devices **1230** radially extending from the stator member **1210**, the one or more stator wellbore engaging devices **1230** contactable with the surface of the wellbore **805**. In accordance with the embodiment shown, the one or more stator wellbore engaging devices **1230** are one or more stator wheels **1235** substantially aligned with a length of the wellbore tractor **1200**. Accordingly, the one or more stator wheels **1235** are configured to substantially prevent rotation of the stator member **1210** relative to the surface of the wellbore **805** as the wellbore tractor **1200** is displaced axially downhole. The phrase “substantially prevent rotation,” as that phrase is used herein, means that the stator rotates at a rate less than 10 percent of a rate of rotation of the turbine **820**.

Turning now to FIG. **13**, illustrated is another wellbore tractor **1300** manufactured and designed according to another embodiment. The wellbore tractor **1300** is similar in many respects to the wellbore tractor **1000** illustrated in FIG. **10**. The wellbore tractor **1300**, in the embodiment of FIG. **13**, employs the base member **810**, one or more turbines **820** and one or more wellbore engaging devices **830** as features of a flow based drive section **1305**. The wellbore tractor **1300**, in addition to the flow based drive section **1305**, includes a powered drive section **1310** coupled to the base member **810**. In accordance with the embodiment shown, the powered drive section **1310** includes one or more powered wellbore engaging devices **1330** radially extending therefrom. In this embodiment, the one or more powered wellbore engaging devices **1330** are also contactable with the surface of the wellbore **805** for displacing the wellbore tractor axially downhole.

In certain embodiments, the flow based drive section **1305** is a primary drive section and the powered drive section **1310** is a secondary hydraulically powered drive section. For example, the secondary hydraulically powered drive section may be designed to displace the wellbore tractor **1300** axially downhole if the primary flow based drive section is unable to do so. As discussed in greater detail

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above, the secondary hydraulic powered drive section may be powered by the fluid flow, for example using the turbine **820** or its own turbine **1320**.

In certain other embodiments, the flow based drive section **1305** is the primary drive section and the powered drive section **1310** is a secondary electrically powered drive section configured to displace the wellbore tractor **1300** axially downhole if the primary flow based drive section is unable to do so.

Depending on the design of the wellbore tractor **1300**, the one or more powered wellbore engaging devices **1330** are one or more powered wheels **1335** positioned at a first powered tilted direction relative to an axial surface of the wellbore **805**. Accordingly, the powered wellbore engaging devices **1330** may also be used to displace the wellbore tractor **1300** axially downhole.

Turning to FIG. **14**, illustrated is another embodiment of a wellbore tractor **1400** manufactured and designed according to the disclosure. The wellbore tractor **1400** shares many of the same features as the wellbore tractor **1300**. The wellbore tractor **1400** primarily differs from the wellbore tractor **1300** in that its powered wellbore engaging devices **1430**, and in the embodiment shown the powered wheels **1435**, are substantially aligned with a length of the wellbore tractor **1300** for displacing the wellbore tractor axially downhole. In this embodiment, the one or more powered wellbore engaging devices **1430** could be movable from a first radially retracted state to a second radially extended state in contact with the surface of the wellbore **805**, as shown by arrow **1440**. The radial movement of the powered wellbore engaging devices **1430** allows them to engage and disengage from the wellbore **805**, such that the flow based drive section **1305** may operate.

Turning to FIG. **15**, illustrated is another, substantially different, embodiment of a wellbore tractor **1500** manufactured and designed according to the disclosure. The wellbore tractor **1500** includes a base member **1510**, having a hydraulically powered drive section **1540** coupled thereto. The wellbore tractor **1500** illustrated in FIG. **15** additionally includes one or more turbines **1520** coupled to the hydraulically powered drive section **1540**. In accordance with this embodiment, the one or more turbines **1520** power the hydraulically powered drive section **1540** based upon fluid **1595** flow across the one or more turbines **1520**, and rotation **1590** thereof. The wellbore tractor **1500** additionally includes one or more wellbore engaging devices **1530** radially extending from the hydraulically powered drive section **1540**, the one or more wellbore engaging devices **1530** contactable with a surface of a wellbore **1505** for displacing the wellbore tractor **1500** axially downhole. In the embodiment shown, the one or more wellbore engaging devices **1530** are one or more powered wheels **1535** that are substantially aligned with a length of the wellbore tractor **1500** for displacing the wellbore tractor **1500** axially downhole. In a simple form of this embodiment, the one or more turbines **1520** power the hydraulically powered drive section **1540** using the fluid **1595**, the hydraulically powered drive section **1540** then being used to displace the wellbore tractor **1500** axially downhole.

Turning now to FIG. **16**, illustrated is yet another embodiment of a wellbore tractor **1600** manufactured and designed according to the disclosure. The wellbore tractor **1600** shares many of the same features as the wellbore tractor **1500** illustrated in FIG. **15**, thus similar reference numerals may be used to indicated similar, if not identical, features. The wellbore tractor **1600** mainly differs from the wellbore tractor **1500**, in that the hydraulically powered drive section

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1540 is a secondary hydraulically powered drive section and the one or more wellbore engaging devices 1530 are one or more hydraulically powered wellbore engaging devices, and further that the wellbore tractor 1600 additionally includes a primary mechanical drive section 1605 coupled to the base member 1510. The primary mechanical drive section 1605, in the illustrated embodiment, includes one or more mechanical wellbore engaging devices 1630 radially extending from the base member 1510. In accordance with this embodiment, the one or more mechanical wellbore engaging devices 1630 are also contactable with the surface of a wellbore 1505 for displacing the wellbore tractor 1600 axially downhole.

The wellbore tractor 1600 illustrated in FIG. 16, in certain embodiments, may include a slip clutch 1650 positioned on the base member 1510 between the one or more turbines 1520 and the hydraulic powered drive section 1540. In accordance with this embodiment, the slip clutch 1650 is configured to fix the one or more turbines 1520 to the base member 1510 and thus displace the wellbore tractor 1600 axially downhole using the one or more mechanical wellbore engaging devices 1630 when in a gripping clutch position. However, the slip clutch 1650 is additionally configured to allow the one or more turbines 1520 to slip with regard to the base member 1510 to power the hydraulically powered drive section 1540 thus displacing the wellbore tractor 1600 axially downhole using the one or more hydraulically powered wellbore engaging devices 1530 when in a slipping clutch position. Additional detail for the slip clutch may be found in above paragraphs.

Turning now to FIG. 17, illustrated is another embodiment of a wellbore tractor 1700 manufactured and designed according to the disclosure. The wellbore tractor 1700 shares many of the same features as the wellbore tractor 1600. The wellbore tractor 1700 further includes an electrically powered drive section 1740 coupled to the base member 1510. In this embodiment, one or more electrically powered wellbore engaging devices 1730 radially extending from the electrically powered drive section 1740. As illustrated, the one or more wellbore engaging devices 1730 are contactable with the surface of the wellbore 1505 for displacing the wellbore tractor 1700 axially downhole.

Further to this embodiment, the slip clutch 1650 is a first slip clutch, and the wellbore tractor 1700 further includes a second slip clutch 1750 positioned on the base member 1510. The second slip clutch 1750, in this embodiment, is configured to fix the one or more turbines 1520 to the base member 1510 and thus displace the wellbore tractor 1700 axially downhole using the one or more mechanical wellbore engaging devices 1630 when in a second gripping clutch position, and configured to allow the one or more turbines 1520 to slip with regard to the base member 1510 to power the electrically powered drive section 1740 thus displacing the wellbore tractor 1700 axially downhole using the one or more electrically powered wellbore engaging devices 1730 when in a second slipping clutch position.

The wellbore tractor 1700 illustrated in FIG. 17, in certain embodiments, further includes one or more second turbines 1720. In this embodiment, the second slip clutch 1750 is positioned on the base member 1510 between the one or more second turbines 1720 and the electrically powered drive section 1740. Accordingly, the first slip clutch 1650 may be used in the conjunction with the hydraulically powered drive section 1540, and the second slip clutch 1750 may be used in conjunction with the electrically powered drive section 1740.

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Those skilled in the art to which this application relates will appreciate that other and further additions, deletions, substitutions and modifications may be made to the described embodiments.

What is claimed is:

1. A method for operating a well system, comprising: placing a wellbore tractor within a wellbore, the wellbore tractor including:

a base member;

one or more turbines fixed to the base member for rotating the base member in a first rotational direction based upon a flow of wellbore fluid in a first direction; and

one or more wellbore engaging devices radially extending from the base member and configured to rotate along with the base member, the one or more wellbore engaging devices contactable with a surface of a wellbore for displacing the base member and one or more turbines axially downhole as the one or more turbines rotate in the first rotational direction; and subjecting the wellbore tractor to a flow of wellbore fluid in the first direction to displace the wellbore tractor axially downhole.

2. The method as recited in claim 1, wherein subjecting the wellbore tractor to the flow of wellbore fluid includes controlling the flow of wellbore fluid from a surface of the wellbore.

3. The method as recited in claim 1, wherein subjecting the wellbore tractor to the flow of wellbore fluid further includes controlling a velocity of the flow of wellbore fluid from the surface of the wellbore to speed up or slow down the displacement of the base member and one or more turbines axially downhole.

4. The method as recited in claim 1, wherein subjecting the wellbore tractor to the flow of wellbore fluid further includes increasing a velocity of the flow of wellbore fluid to a value sufficient to overcome friction between the one or more wellbore engaging devices and the surface of the wellbore and thus push the wellbore tractor uphole.

5. The method as recited in claim 1, further including subjecting the wellbore tractor to a flow of wellbore fluid in a second opposite direction to rotate the base member in a second opposite rotational direction to displace the wellbore tractor axially uphole.

6. The method as recited in claim 1, wherein the base member, one or more turbines and one or more wellbore engaging devices form at least a portion of a flow based drive section, the wellbore tractor additionally including an automation section for performing a downhole task, and further wherein subjecting the wellbore tractor to the flow of wellbore fluid to displace the wellbore tractor axially downhole includes positioning the automation section axially downhole.

7. The method as recited in claim 6, wherein the automation section is a logging tool, and further wherein positioning the automation section axially downhole includes logging downhole wellbore conditions using the logging tool.

8. The method as recited in claim 7, further including releasing the logging tool from at least a portion of the wellbore tractor, thereby allowing the logging tool to return uphole using the flow of wellbore fluid.

9. The method as recited in claim 6, wherein the automation section includes memory and a transmitter, and further wherein positioning the automation section axially downhole includes transmitting information to a downhole device.

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10. The method as recited in claim 6, wherein the automation section includes memory and a transceiver, and further wherein positioning the automation section axially downhole includes receiving information from one downhole device and transmitting the information to another downhole device.

11. The method as recited in claim 6, wherein the automation section is a perforator tool, and further wherein positioning the automation section axially downhole includes perforating the wellbore using the perforator tool.

12. The method as recited in claim 6, wherein the automation section is a sleeve shifting tool having a profile configured to engage with a corresponding profile in a downhole sleeve, and further wherein positioning the automation section axially downhole includes shifting a downhole sleeve.

13. The method as recited in claim 6, wherein the automation section is a swellable packer tool, and further wherein positioning the automation section axially downhole includes deploying the swellable packer tool downhole.

14. The method as recited in claim 1, wherein the wellbore tractor is coupled proximate a downhole end of a

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wireline, and further wherein positioning the automation section axially downhole includes pulling the downhole end of the wireline axially downhole.

15. The method as recited in claim 14, wherein the wellbore tractor is a first wellbore tractor, and further including a second wellbore tractor coupled to an intermediate location of the wireline uphole of the first wellbore tractor, the second wellbore tractor pulling the intermediate location of the wireline axially downhole.

16. The method as recited in claim 1, further including dissolving at least a portion of the wellbore tractor thereby allowing the at least a portion to return uphole after initially subjecting the wellbore tractor to the flow of wellbore fluid in the first direction.

17. The method as recited in claim 1, wherein the wellbore tractor further includes a chute coupled to at least a portion of the wellbore tractor, and further including deploying the chute thereby allowing the at least a portion to return uphole after initially subjecting the wellbore tractor to the flow of wellbore fluid in the first direction.

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